

Department Magazine



Department of
Electronics and **C**ommunication **E**ngineering



Kalasalingam University

October 2012



Volume 4

Issue 7

lectrocomm

find yourself...



From the Editor's Pen

Dear Readers,

As being the Head of the Department of ECE and Chief Editor of this magazine, I feel privileged in presenting the **Seventh Issue** of our Department magazine **Electrocomm**. The purpose of the magazine is to provide a platform to the students and staff to give expression to their original thoughts and thus develop an important aspect of their personality. "**Learning is a continuous process from the minute we are born, until we die**". In recent years, when our students are concerned more about their academic excellence they have to be coaxed, persuaded and encouraged to exhibit their writing talents. The Magazine Team has done therefore its responsibility neatly and systematically, besides of course, editing the magazine. I congratulate the members of the Magazine Team for their tireless efforts. I also send my warm congratulations to all our budding talents who have generously contributed to this magazine.

I extend best wishes for the success of this endeavor!

Dr.S.Durairaj
Editor-in-Chief

Magazine Team Members



Individually, we are one drop. Together we are an ocean.
– Ryunosuke Satoro



KALASALINGAM UNIVERSITY

DEPARTMENT OF ELECTRONICS

&

COMMUNICATION ENGINEERING

ELECTROCOMM

Issue- 7

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“Coming Together is a *BEGINNING*,

Keeping Together is a *PROGRESS*,

Working Together is *SUCCESS*”

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Message

“Learning gives creativity
Creativity leads to thinking
Thinking provides knowledge
Knowledge makes you great”

-Dr.A.P.J.Abdul Kalam

I am very happy to know that the department of Electronics and Communication Engineering is releasing the department magazine **Electro Comm** (issue:7) during the 1st week of October 2012.

I am sure every article in this compilation will be unique and every page will be interesting and inspiring in its own way. This is nothing but a team work with dedication, determination and discipline that leads to success of this release of magazine.

On this occasion, I extend my greetings to all those associated with the magazine and wish them all success in their endeavour and hope that the magazine will prove to be a prized possession for the students and department.

Dr.M.Jeyakumaran

Professor and Head of the Department
Department of Business Administration

MESSAGE

I appreciate the efforts taken by the Department of Electronics and Communication Engineering, Kalasalingam University in releasing the 7th issue of the Department Magazine *ElectroComm*. I am sure the contents of this magazine will motivate all the students of the Department to perform better in order to have a better career.

Every student must have the commitment to learn and be more knowledgeable to succeed. Passion to learn, implement new technologies, and develop better products/gadgets would make the students more competitive.

Communication technologies have advanced to a great extent and there are various modes of communication. Whatever be the mode of communication, students must be able to communicate and express their thoughts clearly, and must also be able to demonstrate their technical capabilities.

Magazines like this will help the student in knowing the latest technologies in Electronics and Communication Engineering and some general information. Best wishes for the release of this magazine.

Prof. Sudhakar Gummadi
Dean – Computing Services &
Head, Dept. of Information Technology

To:

HOD/ECE


18/9/2012



MESSAGE



It gives immense pleasure to know that the department of ECE is bringing forth the 7th issue of their department magazine 'Electrocomm'. It is heartening to see that the students have taken time and pains in their Endeavour in bringing out an attractive, informative and useful magazine that is full of contributions from a broad spectrum of the student community. It is a very good platform given by the department to their students to showcase their skill, talent and multi-faceted personality. I am sure the students would have surprised many of their faculty member and above all themselves by the hidden talents which were dormant all these days. The magazine also proudly displays the list of achievers. This will for ensure stimulate and inspire other students also put an extra effort to see to that their names also find place in such lists.

I wish on and all who have directly or indirectly, actively or passively taken part in this Endeavour good luck and grand success in all walks of life.

Dr.K.Valarmathi

(HoD/ICE)

Kalasalingam University
(Kalasalingam Academy of Research and Education)
Department of ECE

The following are the notable activities in the ECE department, from January 2012 to April 2012.

DATE	NAME OF THE EVENT	RESOURCE PERSONS
Jan 7 2012	Guest Lecture on “ Embedded Systems- The Future ”	Mr.V.Krishnamurthy CEO,EI Labs,Bangalore.
Jan 12 2012	Awareness Program on “ Health Care and Its Importance ” & “Eye Camp” by Vasan Eye Care, Madurai.	- Dr.J.Jeba Singh,Consultant Medical Oncologist, Madurai - Dr.Helan Jeba Singh,Consultant in OBGYN, Madurai. - Dr.S.Ajith Chandrasekar,Director of Vasan Eye Care,Madurai - Chief Guest :Er.S.Sakthivel,Deputy Superintendant of police,Srivilliputtur.
Jan 24 &25 2012	National Conference on “VLSI,Embedded and Communication Systems”(NCVEC 2012)	Invited talks - Dr.Lawrence Jenkins,Professor,IISC Bangalore - Dr.P.G.Krishna Mohan, Professor,JNTU,Hyderabad. - Dr.S.Varada Rajan,Professor,Sri Venkateswara University,Tirupati. - Mr.R.Janakan Rajendran,Global net,USA. - Dr. Soundarapandian, Gandhigram Rural Institute, Dindugal.
Jan 31 2012	Guest Lecture on “Embedded Systems applications”.	Mr.Pankaj Bande, Lead tech consultant, Purple leap, Bangalore.



Feb 28 & 29 2012	Two day National Workshop on “Advanced FPGA DESIGN”	• Mr.S.P.Ramanathan,& Mr.A.David,ENIXS Technology India Pvt Ltd,Trichy.
Feb 29 2012	Placement training for “Non-placed students”	• Mr.S.S.Manian,HRD & Placement trainer, Madurai.
March 27 2012	One day Workshop on” LATEX, LINUX,& PYTHON”	• Mohammed Kasim Khan, Promoter-Self Workshops, Madurai.
March 31 2012	DRDO&BRNS Sponsored One Day National Workshop On “Network Simulation Using NS-2”	• Dr.C.Tharani,Abdur Rahman University,Chennai. • Mrs.G.Sankaramalliga, Vels University,Chennai. • Mr.J.Bhuvanesh kumar,Honey well Tech,Madurai.
April 5 2012	Tech day, Magazine(6th issue)& Milestone Release.	• Dr.SM.Kannan,Principal. ULTRA College of Engg& Tech for Women,Madurai. • Mrs.Madhumithaa,Writer. • Dr.N.V.N.Nampoothiri Professor,Dept of Civil Engg,KLU.
Jan 10 2012	External Counseling	• Sri X.P.White, External Counselor cum Writer, Madurai.
Feb 2 2012		
March 2 2012		
April 3 2012		

The following are the notable activities in the ECE department, from July 2012 to September 2012.

DATE	NAME OF THE EVENT	RESOURCE PERSONS
July 19 2012	Guest Lecture on “Wireless ad-hoc Networks and its Applications”	Mr.Narendran Rajagopalan, Research Scholar, Department of CSE, NIT, Trichy.
July 26 2012	Inaugural function of ASELCOME'12	Dr.R.Ilango, Director of Distance Education, Svyasa University, Bangalore.
Aug 17 &18 2012	National Seminar and Awareness Programme on “Applications Of Radioisotopes and Radiation Technology in Industry and HealthCare”	- Dr. A. K. Kohli, Chief Executive, BRIT - Shri S.K. Malhotra, DAE - Shri.Stanislaus F. D'Souza, Assoc. Director, BARC - Shri.N.Sivaprasad, Sr. Gen Manager, BRIT - Shri.K S S Sarma ,Head EBPS/RTDD BARC & Gen. Manager, BARC - Shri R. I. Bakhtsingh , Electron Beam Centre, BARC - Dr. K. C. Mittal, Electron Beam Centre, BARC - Shri L.N. Bandi, BARC - Shri Amit Sen, AERB
Aug 18 2012	Guest Lecture on “C Programming & Data Structure”	Mr.S.Prasanna, Assistant Professor, Department of Computer Science & Engineering, Thiagarajar college of Engineering, Madurai.
Aug 22 2012	One day Workshop on “Networking of Real CISCO devices”	- Er.P.Athideera Pandian, Director, Pandian Systems and Solutions. - Mr. N. Saravanel, MBA, Pandian Systems and Solutions.
Aug 23 2012	Guest Lecture on “Electronics in Green Energy”	Er. S. Sathanandham, Retd. Deputy General Manager, TNPL.
Aug 30 2012	Guest Lecture on “Radio& Satellite Communication”	Dr.Indumathy, Assistant Professor, MEPCO-SCHLENK Engineering College.



	Guest Lecture on “ Optical Communication ”	• Dr. S. Tamil Selvi, Assistant Professor,National Engineering College.
Aug 31 2012	One day Workshop on “ Applications Development Using JAVA ”	• Ms.V.Amirthamanickam & Ms.K.Karthika HCL CDC,Coimbatore.
Sep 5 2012	Guest Lecture on “ Job Interview& preparation of CV ”	• Mrs.Maya Ethiraj,HRM,Jobs Café Pvt,ltd.
Sep 11 2012	One day Workshop on” LATEX, LINUX,& PYTHON ”	• Mohammed Kasim Khan, Promoter-Self Workshops, Madurai.
Sep 26-28 2012	Guest Lecture on “ Soft skill Training Program ”	• Mrs. S.Abirami, Soft skill trainer, GOOD MORNING (Flourish your skills)
Oct 1 2012	Wii – Pressenza & Magazine Release (7th Issue)	• Dr. S. Jeya Pragasam, Director, International Gandhian Institute for Nonviolence and Peace (IGINP), CESC & Rajarajan Institute of Science (RISE).
July 24 2012	External Counseling	• Sri X.P.White, External Counselor cum Writer,Madurai.
Aug 28 2012		
Sep 25 2012		
July 14 2012	Parents meeting	
Aug 10 2012		
Sep 7 2012		

STAFF ACHIEVEMENTS (2011 - 2012)

Sl.No.	Name of the staff	Conferences		Journal
		National	International	
1	Dr.S.Durairaj	2	2	3
2	Dr. V.K.Madan	2		
3	Dr.T.Arivoli	1	1	
4	Mr. P. Murugan	1		
5	Mr.T.Senthil	1		
6	Mr. P.Sivakumar	2	3	2
7	Mr B.Perumal	2	1	
8	Mr A.Muthukumar	4	3	1
9	Mrs A.Lakshmi	3	2	
10	Mr.B.Srinivasa Rao	1		
11	Mr K.Pandiaraj	2		
12	Mr.S. Sriram Sundar	2		
13	Mrs T. Anitha	3		
14	Ms.N. Sivasankari	5	4	2
15	Mr K. Jeya Prakash	1		
16	Ms.T.Preethiya	4	2	1
17	Ms R.Sumathy	4		
18	Mr G. Karthy	1		
19	Mr. P.Manikandan	2		
20	Mr.M.Suresh	3		
21	Mr.Jenyfal Sampson	1		
22	Ms.G.Ramya Devi	2		
23	Ms.T.Vijaybharathi	3		
24	Ms.Brenda Mohan	1	1	
25	Mr.R. Radeep Krishna	1		
26	Mrs.K.S Dhanalakshmi	2		1
27	Mr.A. Azhagu Jaisudhan Pazhani	1		
28	Mr.J. Prabin Jose	1	1	
29	Mr.S.P. Velmurugan	1		
30	Mr.S.Diwakaran	2		
31	Mr.V.Karutha Raja	2	1	
32	Ms.A.Saraswathi	2		
33	Mrs.M.Arulthenmathi	2	1	
34	Mr.S. Balamurugan	1		
35	Mr.M.Krishna Paramathma	1	1	



FDP Courses/ Workshops Attended :

Sl. no .	NAME OF THE STAFF	FDP COURSES/ WORKSHOPS ATTENDED
1	Dr.S.Durairaj	1) Participated in "Indo-US Interdisciplinary symposium on Issues related to the empowerment of Deaf people" , AYJNIHH, Mumbai , during 7 - 8 August 2012 2) Participated in "National Board of Accreditation workshop on "Training of Master Trainers" , August 31 - September 2 2012, Bhubaneshwar, Orissa.
2	Dr. V.K.Madan	1) Workshop on Advanced Research and International Collaboration on March 23, 2012 at HEIG-VD University, Switzerland and gave an invited talk on "PET Instrumentation". 2) Seminar on Positron Emission Tomography: Technology and Applications, July 30, 2011 at n-CARDMATH, Kalasalingam University, and gave a talk on "PET Instrumentation and Image Reconstruction."
3	Mr. P.Sivakumar	1) 5 Days Training Wipro Mission 10x Programme on UTLP – 10 – 14 September 2012.
4	Mr B.Perumal	1) Participated in National Seminar on"Research and Consultancy Services" Kalasalingam University, 30th-December 2011 2) Participated in "Twoday Workshop on Recent Trends in Medical Image Diagnostics"VIT,Vellore, 25and 26 May, 2012
5	Mr A.Muthukumar	1) Participated in a Workshop on "Signal Processing" at Kalasalingam University held during 17 th September 2011. 2) Participated in a National Seminar on "Research and Consultancy Services" at Kalasalingam University held during 30 th December 2011.
6	Mr.S. Sriram Sundar	1) Advanced FPGA Design Enixs Technology and Kalasalingam University 28 - 29 Feb, 2012 2) Wipro Mission 10x Learners Workshop Wipro 19 – 23 Sep, 2011 3) Wipro Mission 10x Advanced workshop Wipro 15 – 16 Dec, 2011 4) Implementation of Signal Processing Application on FPGA using Xilinx system generator Core EL Technologies and Karunya University, Coimbatore 15 -16 Sep, 2011
7	Ms.N. Sivasankari	1) Two Days workshop on "Advanced FPGA Design Enixs Technology" 28 - 29 Feb, 2012,organized by Kalasalingam University. 2) One day National seminar on "Research and Consultancy

		Services “ on 30.12.2011 at KLU
8	Mr K. JeyaPrakash	1) 5 Days Training Wipro Mission 10x Programme on UTLP – 10 – 14 September 2012
9	Ms.T.Preethiya	1) Participated in Seminar on Wireless Networks at Kalasalingam University on 6 th Sep,2011 2) Two Days workshop on “Simulation of Wireless sensor using NS-2 & Qualnet Networks Simulators” at Mepco Schlenk Engineering College on 5 - 6 December,2011 3) One day National seminar on “Research and Consultancy Services “at Kalasalingam University on 30.12.2011 4) One Day workshop on “Wireless sensor Networks Simulators” at Kalasalingam University on 31.3.2012 5) One Day workshop on “Networking in real time CISCO devices” at Kalasalingam University on 28.8.2012
10	Ms R.Sumathy	1) Three days workshop on “Advanced refresher programme for VLSI faculty with hands on training on Analog mixed signal and digital design using cadence tools” conducted by ICT academy of Tamil Nadu held at PSYEC. 2) One day National seminar on “Wireless Networks” on 6 th September,2011 at KLU.
11	Mr G. Karthy	1) Attended A Two days work shop on “Advanced FPGA Design” held on 28 th and 29 th February at Kalasalingam University,Krishnankoil.
12	Ms.Brenda Mohan	1) Workshop on 'CRYPTOGRAPHY IN VLSI AND EMBEDDED PROGRAMMING' on 24-09-2011 in Sun college of Engineering & Technology. 2) participated in orientation training from 5-03-2012 to 10-03-2012 in Avinashilingam University, Coimbatore 3) National workshop on Network simulation using NS-2 on 31-03-2012 in KLU 4) Workshop on Advanced image processing in MATLAB on 17-08-2012 in Sun college of Engineering & Technology
13	Mrs.K.S. Dhanalakshmi	1) Participated in DRDO &BRNS Sponsored one day National Workshop on “Network Simulation Using NS-2” Organized by Dept. of ECE, KLU
14	Mr.M.Krishna Paramathma	1) National Seminar on " RESEARCH METHODOLOGY" Held on 21 st July 2012 at KLU.

STUDENT'S ACHIEVEMENTS



Mr. Sheik Abdullah from **Department of Electronics and Communication Engineering**, Kalasalingam University, has designed a stethoscope kind of system using the fundamental science principle of '**Compressed Air Mechanism**' just for Rs 30. Using simple air pipes, he has made this device with an amplifier using a small piston that compresses air to amplify sound. The hearing-impaired can fix this tool with mobile phones and attach the ear plugs to hear music as well as use mobile phones like ordinary persons. He was the student of **2008 - 2012 batch**.



Mr. M. Dhamotharan of Third Year, ECE participated and won the **Gold Medal** in the **4th State level Open Yoga Championship – 2012** organized by Athma Pidha Yoga Natural Living Gurugulam held at Tiruchendur on 26th August 2012.

BEST WISHES TO THE WINNERS !!!



Previous Issues of Magazine - A Glance

ELECTROCOMM: 1st Issue on 30th Oct'09



The first issue of the magazine ELECTROCOMM was released by Mr. K. Venkataramanan, ITS, Deputy Manager, BSNL, Virudhunagar Telecom and it was received by Prof.S.Raju, NSS Coordinator (Retd.), Virudhunagar.

ELECTROCOMM: 2nd Issue on 22nd April'10

The second issue of the magazine ELECTROCOMM was released by Dr.P.Subbaraj, Principal, Theni Kammavar Sangam Institute of Technology, Theni and received by Dr. Muthuchezhian, Madurai Kamaraj University



ELECTROCOMM: 3rd Issue on 23rd Sep'10



The third issue of the magazine ELECTROCOMM was released by Mr.S.Annamalai, News Editor, THE HINDU, Madurai and it was received by Mr.A.Ashok, Serial Entertainer, SUN TV.



ELECTROCOMM: 4th Issue on 8th FEB'11

The fourth issue of the magazine ELECTROCOMM was released by Dr.Mrs.R.Sukanesh, Professor, Department of ECE Thiagarajar College of Engineering and was received by Dr. Venkumar, Professor, Dept of Mechanical Engineering, KLU.



ELECTROCOMM: 5th Issue on 8th Sep'11



The fifth issue of the magazine ELECTROCOMM was released by Dr. S.R.Srikumar, Principal, Kalasalingam Institute of Technology and it was received by Mr.Kovai Magesh, Mr.David Vadivel, Mr. Nagesh Chellakkannu, SUN TV Artiste.

ELECTROCOMM: 6th Issue on 5th April'12

The sixth issue of the magazine ELECTROCOMM was released by Mrs. Madhumitha, Writer and it was received by Dr. Nampoothiri, Professor, Dept of Civil Engineering, KLU.



GEMS OF THE DEPARTMENT

Final Year



Sajag Satayu Prasad Singh
ECE-D
CGPA : 9.48



Chandan Kumar
ECE-B
CGPA : 9.36



Keerthana Priya
ECE-C
CGPA : 9.18

Prefinal Year



Pusarla Swarna Madhuri
ECE-B
CGPA : 9.41



Mritunjay Kumar
ECE- B
CGPA : 9.05



V.Gomathi
ECE-A
CGPA : 8.90

Second Year



K.U. Roshna
ECE-C
CGPA : 9.27



S.V. Vignesh
ECE-D
CGPA : 8.9



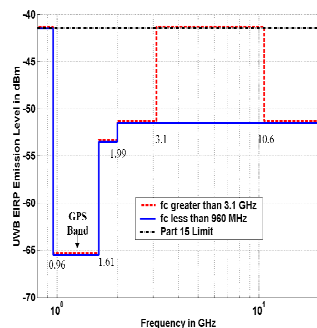
T.Dhuwarahga Shri
ECE-A
CGPA : 8.8

TECHNICAL ARTICLES

Ultra wideband Communications

V.K. Madan

Ultra wideband (UWB) term was coined by U.S. Department of Defense, and presently it characterizes transmission systems with instantaneous spectral occupancy in excess of 500 MHz, or a fractional bandwidth of more than 20%. UWB communication may also be called impulse time-domain, no sinusoidal, baseband, video pulse, ultra high resolution, and carrier less, super wideband communication.

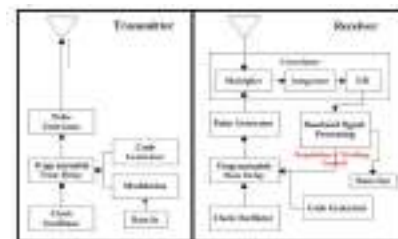


FCC Indoor spectral mask

It is a revolutionary technology with incomparable potential in terms of throughput, performance and low cost implementation, and has attractive features inviting major advances in wireless communications, networking, and radar, imaging, and positioning systems. The United States Federal Communications Commission (FCC) in 2002 was the first

regulatory agency to approve a new wireless UWB communication system and radar, and unleashed huge new bandwidth approximately 3 to 10 GHz at the noise floor with a power levels below -41.3 dBm. Fig. 1 shows FCC spectral mask for indoor communications.

UWB communication systems overlay coexistent RF systems and operate using low-power ultra-short information bearing pulses. Presently products are available like short-range very high-speed broadband access to the Internet, covert communication links, localization at centimeter-level accuracy, high-resolution ground-penetrating radar, through-wall imaging, precision navigation and asset tracking, and a host of applications for health care systems.



An ultra wideband transmitter and receiver

UWB communications rely on nanosecond scale waveforms that can be free of sine-wave carriers and do not require IF processing because they can operate at baseband. UWB radar is an impulse radar or time domain radar. UWB

systems come with unique advantage like enhanced capability to penetrate through obstacles and have small size and processing power and are useful for short-range, high-speed access communication systems, accurate personnel and asset tracking for increased safety and security, precision navigation, imaging of steel reinforcement bars in concrete or pipes hidden inside walls, surveillance, and medical monitoring for example heart's actual contractions.

Modulation Schemes and Applications:

Conventional Narrowband and spread spectrum modulation:

Narrowband modulation schemes include AM radio and analog TV and are sensitive to interfering signal, spike, or noise. The alternative schemes are FM or PM where instantaneous frequency or phase of the carrier is proportional to the modulating signal and are pretty immune to interference signal or noise. More sophisticated narrowband modulation include quadrature amplitude modulation (QAM), and is pioneered in NTSC and PAL-A/B color subcarriers, and is the basis of many cable modems, and with further variations, the HDTV transmission standard.

Shannon's information theory tells us that the ability of a modulation scheme to resist noise and interference depends on the bandwidth. This idea was useful in the invention of spread spectrum communication system. Spread spectrum shares bandwidth between multiple signals concurrently by using different pseudo random (PN) codes for modulation. The codes can be orthogonal like Hadamard codes, and are mutually orthogonal in a cross correlation operation and are the basis of code division multiple accesses (CDMA).

Ultra wideband Modulation:

UWB spreads the bandwidth and exploits Shannon's model to a great degree. UWB unlike narrowband or spread spectrum technique is not carrier based. It is radically different. UWB modulation is based on a very short burst of energy followed by a gap to get a huge spreading ratio. The modulation techniques include singlet impulse, doublets or paired impulses, monocycles and wavelets. The encoding technique includes time modulated encoding scheme, time hopping impulse modulation. Fig. 2 shows a typical UWB transmitter and receiver. The transmitter modulates the narrowband pulse and transmits it. The receiver uses correlator and baseband signal processor to extract the intelligence.

Unlike the conventional modulation schemes, UWB does not suffer from multipath fading. This has advantages for mobile telephony, radar and range finding applications with accuracies under a cm, and a huge range of industrial applications both for commercial and military use.

The biomedical applications include a whole new area for the non invasive measures and monitoring of for example vocal folds, vessels, heart, lung, chest, bladder, and fetus. Prof Enrico Staderini of HEIG-VD is working in this area and is a regular visitor to our university.

UWB technology has progressed greatly. However it has been facing challenges since inception of regulatory issues of bandwidth usage, and acceptance in a marketplace which has never been a hotbed of technological radicalism. Indeed, the lukewarm reception seen by conventional spread spectrum techniques, well understood for decades, would suggest that UWB will see many hurdles yet before it becomes a ubiquitous feature of the technological landscape. However



UWB communications would keep advancing technologically leading to new products with increasing usage.

TELEPORTATION

...Make Your Way through Space

ABSTRACT:

Ever since the wheel was invented more than 5,000 years ago, people have been inventing new ways to travel faster from one point to another. The chariot, bicycle, automobile, airplane and rocket have all been invented to decrease the amount of time we spend getting to our desired destinations. Yet each of these forms of transportation shares the same flaw: They require us to cross a physical distance, which can take anywhere, from minutes to many hours depending on the starting and ending points.

The new millennium has come & gone but still we are using the same old technology for transportation & telecommunication. Now what if you could reach your destination within the blink of an eye, without even crossing a road. What if you could supervise important work in person rather than from your cell phone??What if there were a way to go from home to the supermarket without to use your car or from backyard to the International Space Station without having to board a spacecraft?

There are scientists working right now on such a method of travel by combining properties of telecommunications & transportation to achieve a system called TELEPORTATION.



General process of teleportation

INTRODUCTION:

The word TELEPORTATION was coined in the early 1900s by American fiction writer Charles Fort. Teleportation is the name given by science fiction writers to the feat of making an object or person disintegrate in one place while a perfect replica appears somewhere else. How this is accomplished is usually not explained in detail, but the general idea seems to be that the original object is scanned in such a way as to extract all the information from it, then this information is transmitted to the receiving location and used to construct the replica, not necessarily from the actual material of the original, but perhaps from atoms of the same kinds, arranged in exactly the same pattern as the original. A teleportation machine would work on 3-dimensional objects, it would produce an exact copy rather than an approximate facsimile, and it would destroy the original in the process of scanning it.

Teleportation involves dematerializing an object at one point, and sending the details of that object's precise atomic configuration to another location, where it will be reconstructed. What this means is that time and space could be eliminated from travel -- we

could be transported to any location instantly, without actually crossing a physical distance.

RESEARCH:

In 1993, the idea of teleportation moved out of the realm of science fiction and into the world of theoretical possibility. It was then that physicist Charles Bennett and a team of researchers at IBM confirmed that quantum teleportation was possible, but only if the original object being teleported was destroyed. This revelation, first announced by Bennett at an annual meeting of the American Physical Society in March 1993, was followed by a report on his findings in the March 29, 1993 issue of Physical Review Letters. Since that time, experiments using photons have proven that quantum teleportation is in fact possible.

In 1998, physicists at the California Institute of Technology (Caltech), along with two European groups, turned the IBM ideas into reality by successfully teleporting a photon, a particle of energy that carries light. The Caltech group was able to read the atomic structure of a photon, send this information across 1 meter (3.28 feet) of coaxial cable and create a replica of the photon. As predicted, the original photon no longer existed once the replica was made.

In performing the experiment, the Caltech group was able to get around the Heisenberg Uncertainty Principle, the main barrier for teleportation of objects larger than a photon. This principle states that you cannot simultaneously know the location and the speed of a particle. But if

you can't know the position of a particle, then how can you teleport it? In order to teleport a photon without violating the Heisenberg Principle, the Caltech physicists used a phenomenon known as entanglement.

The most recent successful teleportation experiment took place on October 4, 2006 at the Niels Bohr Institute in Copenhagen, Denmark. Dr. Eugene Polzik and his team teleported information stored in a laser beam into a cloud of atoms. Quantum teleportation holds promise for quantum computing. These experiments are important in developing networks that can distribute quantum information. Professor Samuel Braunstein, of the University of Wales, Bangor, called such a network a "quantum Internet." This technology may be used one day to build a quantum computer that has data transmission rates many times faster than today's most powerful computer.

CLASSIFICATION:

Officially TELEPORTATION has been classified into:

Self-Teleportation:

"The disembodied transport of persons or inanimate objects across space by advanced (futuristic) technological means". This type of teleportation could use a physical connection between the two locations, such as a wire. Using existing technology, such as the Internet, or telephone lines, could be the means for this type of teleport. However, this means it would be extremely slow, since a typical living animal is approximately equal to



600YB (yottabytes). Compression could be utilized, but would have to be effective in keeping all data intact.

P-Teleportation:

"The conveyance of persons or inanimate objects by psychic means."

Vm-Teleportation:

"The conveyance of persons or inanimate objects across space by altering the properties of the space time vacuum, or by altering the space time metric (geometry)." This category includes the use of wormholes for transport, and the modification of the speed of light.

Q-Teleportation:

"The disembodied transport of the quantum state of a system and its correlations across space to another system, where system refers to any single or collective particles of matter or energy such as baryons (protons, neutrons, etc.), leptons (electrons, etc.), photons, atoms, ions, etc."

E-Teleportation:

"the conveyance of persons or inanimate objects by transport through extra space dimensions or parallel universes."

HUMAM TELEPORTATION:

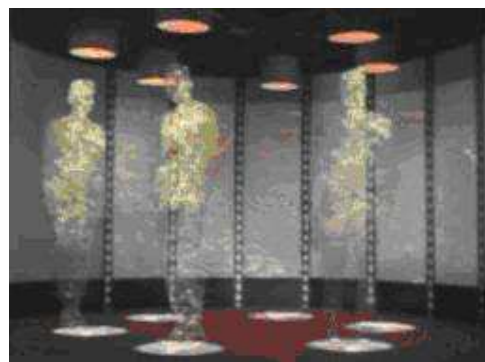
The use of teleportation as a means of transport for humans still has considerable unresolved technical and philosophical issues, such as exactly how to record the human body sufficiently accurately and also be able to reconstruct it, and whether destroying a human in one place and recreating a copy elsewhere

would provide a sufficient experience of continuity of existence. Believers in the supernatural might wonder if the soul is recopied or destroyed, and might even consider it murder. Likewise, someone with a secular worldview who considers the body synonymous with the self might also see the disintegration of a given corpus as the killing of a human being. The reassembled human would be a different sentience with the same memories as the original. Many of the questions are shared with the concept of mind transfer.



The process of teleportation on a human

- **The process of teleportation on a human** - The laws of physics may even make it impossible to create a transporter that enables a person to be sent instantaneously to another location, which would require travel at the speed of light.



Animated view of teleportation

- **Animated view of teleportation-**

For a person to be transported, a machine would have to be built that can pinpoint and analyze all of the 102k atoms that make up the human body. That's more than a trillion atoms. This machine would have to send this information to another location, where the person's body would be reconstructed with exact precision. Molecules couldn't be even a millimeter out of place, lest the person arrive with some severe neurological or physiological defect.

CONCLUSION:

If a teleporter is really built, the subject will be teleported to the destination at the speed of light, which is the real advantage & spirit behind the research of teleportation. So think about your office or college as easily as you were going to house in your neighborhood or reaching a star or a planet at a speed which Newton or Einstein have not dreamed to be possible.

As all good things have limits teleportation also has its limits. The teleporter transfers matter at a speed of light & we know that light has its speed limit; the feeling of instantaneous teleportation is felt only on Earth & distances comparable with, or less than the speed of light. If a subject is transported to a galaxy at distance of about 4 light years away then it will take a period of 4 years (of earth time) for the subject to reach its destination.

As everything invented or discovered in science can be used for or against mankind, this wonderful technology could also be used in an illegal way by trapping people in the form of a

signal, or send destructive materials such as bombs.



A Futuristic view of a teleporter

So this technology which is still in its infancy has real hopes for the future of mankind.

-B.Naga Nandhini

ECE-B

Prefinal year





4 G-TECHNOLOGIES

With the advent of Wireless technology there opens a new door to the communication standard. 4G is the fourth level of the wireless generation. The implementation and access to the 4G network is still in its development stage. All the giants of mobile industry and tech engineers are busy to research and devise it with full confidence. The specification under consideration is the high speed for the data transfer with online security. 4G networks will hopefully come into the field with the promises it is making with the customer of speed and security. 4G network or wireless communication is not a new terminology as 1G, 2G & 3G are already in the market and has been used by the people since 1980. All generations have their particular traits to attract the user for its use. 1G is the first complete wireless standard removes the fuss of wiring and per channel 1 user. The lodging of multi user in one channel becomes possible when 2G comes with improved version after a decade (1990) of 1G. 3G revolutionized the concept of internet. 3G is the 21st generation network and is enjoying its privilege. The successor of 3G is now its way of implementation and is known as 4G. The video facility and speed which is introduced by 3G will be the true essence of 4G. It promises all basic features together with high speed.

WHAT IS 4G MOBILE?

4G can be best described in one word "MAGIC". MAGIC-Mobile Multimedia Anytime Anywhere Global mobility support, Integrated wireless and personalized services. The next generation of wireless technology that promises higher data rates and expanded multimedia services. A completely new fully IP-based

integrated system or systems and network of networks achieved after convergence of wired and wireless networks.

OBJECTIVES OF 4G:

Speeds up to 50 times higher than of 3G. Three-dimensional virtual reality—imagine personal video avatars and realistic holograms. 4G will solve problems like limited bandwidth in 3G. When people are moving and uncertainty about the availability of bandwidth for streaming to all users at all times. Seamless connectivity and global roaming across multiple networks.

PRINCIPLES OF 4G:

- 4G will base on OFDM (Orthogonal Frequency Division Multiplexing).
- MIMO-Multi input And Multi Output
- Adaptive radio interface
- Modulation, spatial processing including multi-antenna and multi-user MIMO.

How 4G Technology Works?

As the need for communication rather fastest communication is the foremost priority of present era also the need of quick data transfer. Distant business correspondence by sharing data becomes very important. Ever growing technology is the example of one such step towards the fastest transmission of data. The 4G stands for fourth Generation is the latest technology with high speed transferability of data with security measurements. It is coming with

wireless broadband for the instant download.

Talking about the standard of 4G technology, still not defined as set standard, two technologies are supposed to be the based features of 4G.

- Wi-MAX
- LTE

ITU promotes the technologies against the defragmentation and incompatibilities in 4G technologies.

Wi-MAX stands for Worldwide Interoperability of Microwave Access previously worked as fixed wireless facility under the 802.16e band. Now the modified standard 802.16m has been developed with the properties of speed, wide spectrum, and increase band.

4G has an advantage of having the WiMAX as a product because IEEE introduces and releases it already therefore economic as no need to pay for its manufacturing price. 4G supports two basic equipments;

- Wi-MAX Network system (network infrastructure)
- mobile phone set

Smart phones with Wireless Access are going to be introduced in the market are the model 4Gmobiles. These Smartphone are equipped with the wireless internet accessibility and no fear of losing connection while travel from one tower to another tower range.

Based on the IP wireless connectivity, it increases the optimization for the internet. It manages the voice through packet-switching instead of circuit switching. Internet connectivity with specific IP not only increases the speed but

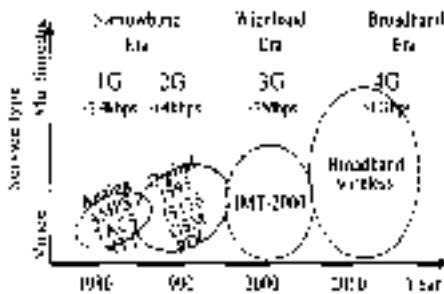
also reliability of the sending and receiving of data.

During a phone call when caller send the information by connecting to WiMAX network, this information first processed to the internet home and then spread widely. Most of the time this transmission happens very fast problems arise in case of spectrum, bandwidth and data. In case spectrum is not wide, shorten bandwidth and specific data carries through the internet. Arrival of 4G has diminished all the fears of lower bandwidth, narrow spectrum and amount of data send / receive. This WiMAX technology has a high speed of data transfer rate with additional capacity for the subscribers and ready to carry big amount of data. Previous generations were suffering because of low speed which ultimately covered in the 4G.

Parallel to Wi MAX, LTE (Long Term Evolution) is introduced by Verizon. LTE is considered to be promising high data transfer speed. LTE is supposed to provide internet facility using both systems. It has the ability of transition from one mode to another. LTE is developed on radio waves technology. This not only increases the speed but also the amount of data allowed through the same bandwidth and results into lower cost.

As LTE is compatible with 3G technology so it not only increases the speed but also prevents the need of new network and can work through the same infrastructure. LTE will not only support the functions of 3G but also incorporate some newer ones. LTE is using MIMO (Multiple input multiple output) able to send and receive huge data negative in the sense that it will overload the base stations networks.

PICTORIAL VIEW OF EVOLUTION OF 4G:



APPLICATIONS OF 4G:

- E-commerce
- Business/Work
- Private Life
- Vehicular
- Public Place
- Entertainment
- Education

-Gautham Siva
ECE-A
Second year

5G (5th generation mobile networks or 5th generation wireless systems)

5G is a technology used in research papers and projects to denote the next major phase of mobile telecommunication standards beyond the 4G/IMT-Advanced standards. 5G is not officially used for any specification or official document yet made public by telecommunication companies or standardization bodies such as 3GPP,

WiMAX Forum, or ITU-R. New standard releases beyond 4G are in progress by standardization bodies, but are at this time not considered as new mobile generation. The 5G family of ITU standards could be implemented around the year 2020.

A new mobile generation has appeared every 10th year since the first 1G system (NMT) was introduced in 1981, including the 2G (GSM) system that started to roll out in 1992, 3G (W-CDMA/FOMA), which appeared in 2001, and "Real 4G" standards which is set at 1000mbps speeds, will only be realized many years into the future. Predecessor technologies have appeared on the market a few years before the new mobile generation, for example the pre-3G system CDMA One/IS95 in 1995, and the pre-4G systems Mobile WiMAX and first release-LTE in 2005 and 2009 respectively.



The development of the 2G (GSM) and 3G (IMT-2000 and UMTS) standards took about 10 years from the official start of the R&D projects, and development of 4G systems started in 2001 or 2002. However, still no transnational 5G development projects have officially been launched, New mobile technologies are generally assigned new frequency bands and wider spectral bandwidth per frequency channel (1G up to 30 kHz, 2G up to 200 kHz, 3G up to 5 MHz frequency bands or larger and industry

representatives have expressed scepticism towards, and 4G up to 40 MHz), but skeptics argue that there is little room for new 5G. Channel bandwidth From users point of view, previous mobile generations have implied substantial increase in peak bit rate (i.e. physical layer net bitrates for short-distance communication). However, no source suggests 5G peak download and upload rates of more than the 1 GBPS to be offered by ITU-R's definition of 4G systems. If 5G appears, and reflects these prognosis, the major difference from a user point of view between 4G and 5G techniques must be something else than increased maximum throughput; for example lower battery consumption, lower outage probability (better coverage), high bit rates in larger portions of the coverage area, cheaper or no traffic fees due to low infrastructure deployment costs, or higher aggregate capacity for many simultaneous users (i.e. higher system level spectral efficiency). Those are the objectives in several of the research papers below.

What is the need for 5G–Technology?

Mobile broadband is becoming a reality, as the internet generation grows accustomed to having broadband access wherever they go and not just at home or



in the office. Of the estimated 3.4 billion people who will have broadband by 2014, about 80 percent will be mobile broadband subscribers – and the majority will be

served by High Speed Packet Access (HSPA) and Long Term Evolution (LTE) networks. There is strong evidence supporting predictions of increased mobile broadband usage. But you might have an uncertainty at this movement (i.e.) all the above requirements are expected to be satisfied by LTE advance itself. Then why there is a need for 5G? Yes of course LTE might support peak data rate of DL: 1Gbps, UL: 500Mbps within a scalable bandwidth where the user can gratify his requirement. In turn LTE advance provides beyond the demand. The actual dilemma starts here.

What makes 5G?

Even though LTE provides wide range of growth for present wireless telecommunication. People are not in a circumstance to make use of those benefits in an effective manner. LTE might be rigorously used in Commercial/Industrial areas. But think of a common man who utmost utilize LTE for downloading a movie or to make a video call. Fact is that there is no such ground-breaking application exists in real world to be utilized by a common man. You might doubt how this verdict is applicable for current innovative world, where we have enormous splendid real time applications. Concern is that our present wireless telecommunications is bottlenecked to use those applications in an effective manner. Our research mainly focuses on how a 5G network can provide more approach to a common man to utilize his available possessions in an immense way to make him to feel the real progress. The 5G Noncore will be a convergence of below mention technologies. These technologies have their own impact on



existing wireless network which makes them in to 5G.

- Nanotechnology.
- Cloud Computing.
- All IP Platform.

CONCLUSION:



Hence, the process of 5G is under construction.....not yet completed.

-Gautham Siva
ECE-A
Second year

SIXTH SENSE TECHNOLOGY

INTRODUCTION:

Sixth Sense Technology is a mini-projector coupled with a camera and a cell phone which acts as the computer and connected to the Cloud, all the information stored on the web. Sixth Sense can also obey hand gestures. The camera recognizes objects around a person instantly, with the micro-projector overlaying the information on any surface, including the object itself or hand. Also

can access or manipulate the information using fingers.

Sixth Sense is a wearable gestural interface that augments the physical world around us with digital information and lets us use natural hand gestures to interact with that information. By using a camera and a tiny projector mounted in a pendant like wearable device. Sixth Sense sees what you see and visually augments any surfaces or objects we are interacting with. It projects information onto surfaces, walls, and physical objects around us, and lets us interact with the projected information through natural hand gestures, arm movements, or our interaction with the object itself. Sixth Sense attempts to free information from its confines by seamlessly integrating it with reality, and thus making the entire world your computer.



Parts used in 6th Sense Technology

HOW SIXTH SENSE TECHNOLOGY WORKS?

- The technology in itself is nothing more than the combination of some stunning technologies, but the idea

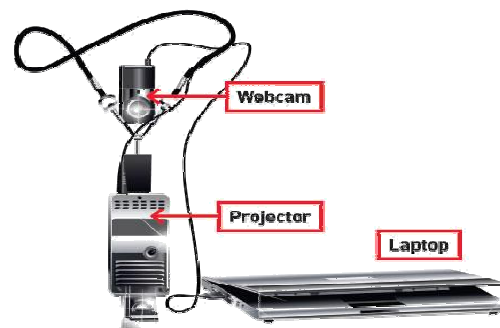
of combining those technologies is really great.

- The technology is mainly based on hand gesture recognition, image capturing, processing, and manipulation, etc.
- The camera is used to recognize and track user's hand gestures and physical objects using computer-vision based techniques, while the projector is used to project visual information on walls or on any physical thing around us. Other hardware includes mirror and colours caps to be used for fingers.
- The projector: projects visual information, enabling surfaces, walls and physical objects around the wearer to be used as interfaces.
- The camera and hands: recognizes and tracks the user's hand gestures and physical objects using computer-vision based techniques.
- The software program: processes the video stream data captured by the camera and tracks the locations of the colored markers (visual tracking of colored fingers) at the tip of the user's fingers using simple computer-vision techniques. The movements and arrangements of these fiducially are interpreted

into gestures that act as interaction instructions for the projected application interfaces.

- The maximum number of tracked fingers is only constrained by the number of unique fiducially, thus Sixth Sense also supports multi-touch and multi-user interaction.

ARCHITECTURE OF SIXTH SENSE TECHNOLOGY:



Architecture of 6th Sense Technology

- Sixth Sense, a tool that connects the physical world with the world of data.
- Digital prototype called the “Sixth Sense”, which wearable hi-tech devices that can turn anything into a touch screen. Your wrist can become a watch and your palm a phone. The device is combination of a wearable camera, a projector and a mirror where the camera recognizes movements of the fingers of your hand and translates



those into commands.

- The hardware components are coupled in a pendant like mobile wearable device. Both the projector and the camera are connected to the mobile computing device in the user's

pocket. The projector projects visual information enabling surfaces, walls and physical objects around us to be used as interfaces; while the camera recognizes and tracks user's hand gestures and physical objects using computer vision based techniques.

- The software program processes the video stream data captured by the camera and tracks the locations of the colored markers (visual tracking fiducially) at the tip of the user's fingers using simple computer vision techniques.
- The movements and arrangements of these fiducially are interpreted into gestures that act as interaction instructions for the projected application interfaces. The maximum number of tracked fingers is only constrained by the number of unique fiducially, thus Sixth Sense also supports multi touch and multi user interaction.

APPLICATIONS OF SIXTH SENSE TECHNOLOGY:

The Sixth Sense prototype implements several applications that demonstrate the usefulness, viability and flexibility of the system.

- The map application lets the user navigate a map displayed on a nearby surface using hand gestures, similar to gestures supported by multi-touch based systems, letting the user zoom in, zoom out or pan using intuitive hand movements.



Map Application

- The drawing application lets the user draw on any surface by tracking the fingertip movements of the user's index finger. Sixth Sense also recognizes user's freehand gestures (postures). For example, it implements a gestural camera that takes photos of the scene the user is looking at by detecting the 'framing' gesture.

The user can stop by any surface or wall and flick through the photos he/she has taken.



Drawing Application

- Sixth Sense also lets the user draw icons or symbols in the air using the movement of the index finger and recognizes those symbols as interaction instructions. For example, drawing a magnifying glass symbol takes the user to the map application or drawing a '@' symbol lets the user check his mail.
- The Sixth Sense system also augments physical objects the user is interacting with by projecting more information about these objects projected on them. For example, a newspaper can show live video news or dynamic information can be provided on a regular piece of paper. The gesture of drawing a circle on the user's wrist projects an analog watch.



Drawing a circle on the user's wrist projects an analog watch.

- Make a call by Extend hand on front of the projector and numbers will appear for to click. Know the time by Draw a circle on wrist and a watch will appear. take a photo by Just make a square with fingers, highlighting what want to frame, and the system will make the photo—which can later organize with the others using own hands over the air.
- And the device has a huge number of applications, it is portable and easily to carry as can wear it in neck. The drawing application lets user draw on any surface by observing the movement of index finger. Mapping can also be done anywhere with the features of zooming in or zooming out. The camera also helps user to take pictures of the scene is viewing and later can arrange them on any surface. Some of the more practical

uses are reading a newspaper. Reading a newspaper and viewing videos instead of the photos in the paper. Or live sports updates while reading the newspaper.



News paper

- The device can also tell arrival, departure or delay time of air plane on tickets.
- For book lovers it is nothing less than a blessing. Open any book and find the Amazon ratings of the book. To add to it, pick any page and the device gives additional information on the text, comments and lot more add on feature.
- However, instead of requiring you to be in front of a big screen like Tom Cruise, Sixth Sense can do its magic and a lot more everywhere, even while you are jumping hysteric over Oprah's sofa. The camera recognizes objects around you instantly, with the micro-projector overlaying the

information on any surface, including the object itself or your hand. Then, you can access or manipulate the information using your fingers.

- Need to know the time? Draw a circle on your wrist and a watch will appear. Want to take a photo? Just make a square with your fingers, highlighting what you want to frame and the system will make the photo which you can later organize with the others using your own hands over the air. But those are just novelty applications.
- Need to make a call? Extend your hand on front of the projector and numbers will appear for you to click.



Numbers on Hand to make a Call

- The true power of Sixth Sense lies on its potential to connect the real world with the Internet, and overlaying the information on the world itself. Imagine you are at the

supermarket, thinking about what brand of soap is better. Or maybe what food you should get for tonight's dinner. Just look at objects, hold them on your hands, and Sixth Sense will show you if it's good or bad, or if it fits your preferences or not.

- Now take this to every aspect of your everyday life. You can be in a taxi going to the airport, and just by taking out your boarding pass, Sixth Sense will grab real time information about your flight and display it over the ticket. You won't need to do any action. Just hold it in front of your and it will work.

The key here is that Sixth Sense recognizes the objects around you, displaying information automatically and letting you access it in any way you want, in the simplest way possible.

Clearly, this has the potential of becoming the ultimate "transparent" user interface for accessing information about everything around us. If they can get rid of the colored finger caps and it ever goes beyond the initial development phase, that is. But as it is now, it may change the way we

interact with the real world and truly give everyone complete awareness of the environment around us.

FUTURE ENHANCEMENT:

Further progress has been made with the system by broadcasting their latest demonstration online. The new Sixth Sense system uses a camera that is hung from the user's neck, instead of mounted on your hat. Also hanging from the neck is a Pico projector and a mirror that will display onto any surface that the user is looking at. When we first heard about this system, the user had to have a mini computer strapped to their back that would always be connected to the internet and could find the required information. The new system eliminates the backpack and instead connects to your cell phone to retrieve information. The Sixth Sense is still in its development stages.

CONCLUSION:

Technology has transformed the world in which we live, changing how we spend our time, how we understand ourselves, and how we interact with others. Technological innovation results in social and economic change. Thus, Sixth Sense Technology will lead to the development of present world. And it is the present world that promises to restructure human life and activity.

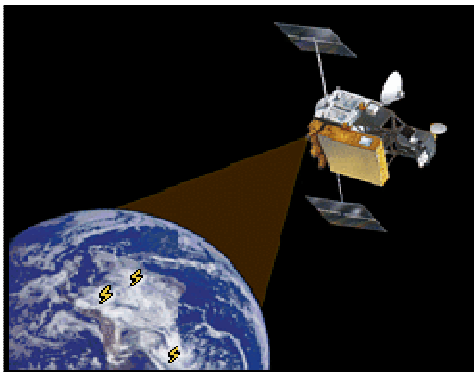
-B.Naga Nandhini
ECE-B
Prefinal year



LIGHTNING IMAGING **SENSOR**

The Lightning Imaging Sensor (LIS), is a space based instrument used to detect the distribution and variability of total lightning (cloud-to-cloud, intracloud, and cloud-to-ground lightning) that occurs in the tropical regions of the globe. The LIS is a science instrument aboard the TRMM Observatory, which was launched on 28 November 1997 from the Tanegashima Space Center in Japan.

This lightning sensor consists of a staring imager who is optimized to locate and detect lightning with storm-scale resolution (4 to 7 km) over a large region (600 x 600 km) of the Earth's surface. The TRMM Satellite travels a distance of 7 kilometers every second (nearly 16,000 miles per hour) as it orbits the Earth, thus allowing the LIS to observe a point on the Earth or a cloud for almost 90 seconds as it passes overhead. Despite the brief duration of an observation, it is long enough to estimate the flashing rate of most storms. The instrument records the time of occurrence, measures the radiant energy, and determines the location of lightning events within its field-of-view.



This calibrated lightning sensor uses a wide field-of-view expanded optics lens with a narrow-band filter in conjunction with a high speed charge-coupled device detection array. A Real Time Event Processor (RTEP), inside the electronics unit, is used to determine when a lightning flash occurs, even in the presence of bright sunlit clouds.

Weak lightning signals that occur during the day are hard to detect because of background illumination. The RTEP will remove the background signal, thus enabling the system to detect weak lightning and achieve 90% detection efficiency. Data from the Lightning Imaging Sensor is being used to study musicale phenomena such as storm convection, dynamics, and microphysics.

The LIS instrument was designed by the GHCC Lightning Team and was manufactured at the Marshall Space Flight Center in Huntsville, Alabama. LIS will contribute significantly to several TRMM mission objectives by providing a global lightning and thunderstorm climatology from which changes (even subtle temperature variations) might be easily detected. The LIS instrument is part of the Tropical Rainfall Measuring Mission (TRMM). TRMM is a joint mission between NASA and the National Space Development Agency (NASDA) of Japan designed to monitor and study tropical rainfall and the associated release of energy that helps to power the global atmospheric circulation shaping both weather and climate around the globe.

LIS EOS Mission Objectives:

- ✓ Cloud Characterization
- ✓ Hydrologic Cycle Studies
- ✓ Storm Convection
- ✓ Microphysics and
- ✓ Dynamics

V.P.KALYANKUMAR
ECE-B
Final year

SWITCHES USED IN HOME APPLIANCES

We are aware of common home appliance switches that perform the operation ON & OFF. But there are some types of switches that are used in Washing Machines, Electric Rice Cookers, and Microwave Ovens and also in Automobiles. These vary with the switching terminology such as rest position, operating point, movement differential, etc.

There are about 140 different types of switches. For example let us consider K4 series switch. These switches are used in Electric Rice Cooker for push operation. It has a flexible actuator made up of Cold-Rolled Steel.



ASSEMBLING OF COMPONENTS:

It is done using “Standard Operating Procedure (SOP)” as per Instructions Catalogue. In this company, they are manufacturing two types.

- (i) E Series Switch
- (ii) K Series Switch

E SERIES SWITCH:

Temperature : -40 to 80 Centigrade.

Flammability rating : UL94VO

Materials

Case: Thermoplastic Polyester

Switch : Thermoplastic Acetyl

Terminals: Coating of Brass is done for CT, whereas NC & NO are not coated

Moving Blade : Beryllium Copper

Spring : Stainless Steel

Contacts: Silver Alloy,

K SERIES SWITCH:

Case / Cover: Thermoplastic Polyester

Switch : Thermoplastic Polyester

Materials

Case: Thermoplastic Polyester

Switch : Thermoplastic Acetyl

Moving Blade : Beryllium Copper

Spring : Stainless Steel

Contacts: Silver Alloy, Silver

Cadmium Oxide, Gold Cross Point

In case for heavy duties, we need to use case and cover made up of Phenol.

BASIS OF SWITCHES:

Switch terminology

1. Rest position
2. Pre-travel
3. Operating point
4. Over travel
5. Movement differential
6. Reset point
7. Reset travel



Based on switch circuitry, switches are divided into six types. Such as:

1. SPDT – Single Pole Double Throw
2. DPDT – Double Pole Double Throw
3. SPSTNC – Single Pole Single Throw Normally Close
4. SPSTNO – Single Pole Single Throw Normally Open
5. DPSTNC – Double Pole Single Throw Normally Close
6. DPSTNO – Double Pole Single Throw Normally Open

PACKING OF SWITCH:

- NC, NO and CT are soldered to separate wires. In order to invisible the soldered joint, we are doing the process of mounding.
- At last, the switch is covered with a rubber boot.
- Switch is operated with a resistance of 40 million Ω
- At the end, the switch is tested with “End of Line Tester (EOL)”

In some switches, such as D4 series type, there are two CT, two NC and two NO terminals. In that case, testing is different. It does not involve any manual operation.

1. Air pressure is given for the movement of cylinders called as, PNEUMATIC CYLINDERS.
2. The mechanical movement is operated by a solenoid.
3. The side cylinder is operated with the help of microprocessor which has the inbuilt memory in the form of C Program.

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
void main()
{
    int a,b,c,d,i;
    char start,cmd;
    clrscr();
    cmd=strcpy(start);
    printf("Enter the command\n");
    scanf("%s",&cmd);
    printf("Process started\n");
    printf("Hold enabled\n");
    printf("Testing\n");
    if(a!=b)
        printf("Fail");
    else if(a==b)
        for(i=0;i<=5;i++)
        {
            if(c==d)
                printf("Pass");
            else
                printf("fail");
        }
    getch();
}
```

Output:

Enter the command
Start
Process started
Hold enabled
Testing
Pass

-N.HARINI

ECE-B

Final year

Life is a boat,
We sail on.
Life is a festival,
We celebrate.
Life is a flower,
We bloom and die.
Life is of God,
He made them all
....Magazine Team

ELECTRONIC CONTACT LENS DISPLAYS PIXELS ON THE EYES



The future of augmented-reality technology is here - as long as you're a rabbit. Bioengineers have placed the first contact lenses containing electronic displays into the eyes of rabbits as a first step on the way to proving they are safe for humans. The bunnies suffered no ill effects, the researchers say.

The first version may only have one pixel, but higher resolution lens displays - like those seen in Terminator - could one day be used as satnav enhancers showing you directional arrows for example, or flash up texts and emails - perhaps even video. In the shorter term, the breakthrough also means people suffering from conditions like diabetes and glaucoma may find they have a novel way to monitor their conditions.

In February, New Scientist revealed the litany of research projects underway in the field of contact lens enhancement. While one company has fielded a contact lens technology using a surface-mounted strain gauge to assess glaucoma risk, none have built in a display, or the lenses needed for focused projection onto the retina - and then tested it in vivo. They have now.

"We have demonstrated the

operation of a contact lens display powered by a remote radiofrequency transmitter in free space and on a live rabbit," says a US and Finnish team led by Babak Praviz of the University of Washington in Seattle.

"This verifies that antennas, radio chips, control circuitry, and micrometer-scale light sources can be integrated into a contact lens and operated on live eyes."



I'll be back: Could electronic contact lenses give humans Terminator-like vision in the future? (Image: Solvents News & Photo Agency/Rex Features)

The test lens was powered remotely using a 5-millimetre-long antenna printed on the lens to receive gigahertz-range radio-frequency energy from a transmitter placed ten centimeters from the rabbit's eye. To focus the light on the rabbit's retina, the contact lens itself was fabricated as a Fresnel lens - in which a series of concentric annular sections is used to generate the ultra short focal length needed.

They found their lens LED glowed brightly up to a meter away from the radio source in free space, but needed to be 2 centimeters away when the lens was placed in a rabbit's eye and the wireless reception was affected by body fluids. All the 40-minute-long tests on live rabbits were performed under general anesthetics



and showed that the display worked well - and fluorescence tests showed no damage or abrasions to the rabbit's eyes after the lenses were removed. While making a higher resolution display is next on their agenda, there are uses for this small one, say the researchers: "A display with a single controllable pixel could be used in gaming, training, or giving warnings to the hearing impaired."

"This is clearly way off in the future. But we're aware of the research that is ongoing in this field and we're watching the technology's potential for biosensing and drug delivery applications in particular," says a spokesperson for the British Contact Lens Association in London.

Contact Lenses with Embedded Electronics:

Scientists at the University of Washington have created contact lenses with embedded electronic circuitry and lights. This development brings us a step closer to lenses that could enable us, for example, to zoom in on distant objects or lenses that can superimpose information on the external environment (augmented reality).

The Experiment & Status:

Although the prototype does not provide any functionality as of yet, the lenses are proof of concept that such devices can be created. Readers will know that they are quite in love with head-mounted display systems and augmented reality now each one also know what they want for their birthday in 2027. Although only if they've managed to have the rabbits wear it for more than 20 minutes without Diode Grid Self-Assembles, Solar Powered.

Looking at the pictures, it appears as if we can count an 8×8 grid makes me wonder what resolution we will have reached by 2020-2030. As it stands, the circuitry is only a few **nanometers** thick (about one thousandth the width of a human hair).

They constructed light-emitting diodes one third of a millimetre across. They then sprinkled the greyish powder of electrical components onto a sheet of flexible plastic.



The shape of each tiny component dictates which piece it can attach to, a micro fabrication technique known as self-assembly. Capillary forces – the same type of forces that make water move up a plant's roots, and that cause the edge of a glass of water to curve upward – pull the pieces into position.

They state in the press release that lenses with a few pixels could be available "fairly quickly", the goal being to power them by a combination of radio-frequency power and solar panels placed on the lens outside- and around the iris.

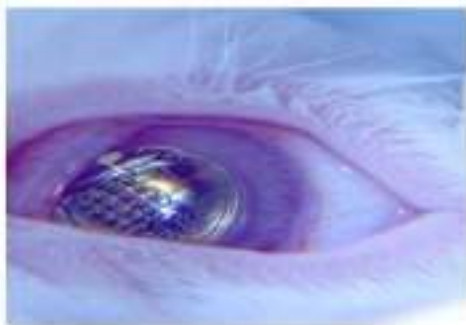
The project was led by Harvey Ho, a former graduate student of [Babak Parviz (electrical engineer at Uni. of Washington)] now working at Sandia National Laboratories in Livermore, Calif., who presented the results at the Institute of Electrical and Electronics Engineers'

international conference on Micro Electro Mechanical Systems in Tucson, AZ.

Future Potential and Related Tech:

We're using our phones to track our position, view maps and browse the web. Google Earth and similar software is populating our external environment with an artificial layer of (Meta) data to increase our productivity and capabilities. Lenses (or glasses) can rid us of the miniature phone screens and project this artificial layer of data directly into our eyes.

The developers themselves mention use cases such as heads-up display for pilots or drivers to show direction and speed. It's not hard to imagine more advanced versions and methods of control; sensors on your arms that track hand gestures and allow interaction with the objects displayed via the lens. Parviz said "We need to improve the antenna design and the associated matching network and optimize the transmission frequency to achieve an overall improvement in the range of wireless power transmission."



"Our next goal, however, is to incorporate some predetermined text in the contact lens."

-ROSHNA.K.U.
ECE-C
Second year

HOME AUTOMATION

What is the mean of home automation system?

Automation is the process of automatically performing everyday functions around the home to save you time, energy, money and at the same time offering improved security.

Why a home automation system?

To satisfy the need to increase:

- Health (reaction on a CO detection)
- Safety (good lighting to avoid falling)
- Security (burglar detection)
- Indoor environment (increase air quality)
- Sustainability (reduce energy consumption)
- Comfort (adjusting air temperature)

Block Diagram:



What are the types of home automation?

Individual control devices:

This type is perhaps the first one to hit the market in the early years. Here each device like the heater or your AC will have an independent control dedicated for it



unlike the central control system. These are very handy and are user friendly.

Distributed Control Systems:

The main feature of this type of systems is emergency shut-down. Rest of the features includes hardware and software components developed under SCADA (special software) for better control automated control that reduce human interaction. With this you can preset or change the control parameters, for example, the thermostat of several ACs and ON/OFF timings.

Centrally Controlled Systems:

Central control is the heart of all automation system. These are computerized systems programmed to handle all functions of multiple utilities like ACs, heaters, home entertainments, doors, windows, refrigerators and cooking systems etc regardless of whether you are home or out. You can connect to your control system from office through telephone or internet. The latest information on home automation system can be had from the suppliers who install it. The best part of this is, with computerized control, you can afford to forget things and still be safe and timely on controls.

10 Important application of home automation:

- > Lighting
- > Security System
- > HI-FI (HAI s/m)
- > Internet Connected video camera
- > Monitor & Program
- > Thermostat
- > Telephones
- > Motion Detection
- > Vehicle Detection

- > Home Entertainment s/m
- > Gas and Electric Fire places
- > Watering System

MobiLinc Lite Insteon and X 10 Controllers:

Using Wi-Fi, or cellular communication, MobiLinc supports control and real-time monitoring of up to 1024 INSTEON or X10 devices, 1024 scenes, and 1024 programs on your Android powered device anywhere in the world. With MobiLinc Pro and Android, control all of your lights, thermostats, devices, programs and irrigation system while on vacation or away from home. With MobiLinc and the many INSTEON devices available, home automation from the palm of your hand has never been easier. MobiLinc Pro works with the ISY-99i central controllers or the SmartLinc.

- Command Your Devices Anywhere with Wi-Fi or cellular access
- Automatic Find and Sync with ISY and SmartLinc on Local Wi-Fi
- Real-Time INSTEON Device and Scene Status
- Secure Connect to ISY over HTTPS using Username and Password Authentication
- Supports up to 1024 INSTEON Devices, 1024 Scenes, and 1024 Programs
- Multiple Profile Support
- Keep Your Favorite Devices, Scenes, and Programs on a dedicated page
- Supports Multiple Thermostats
- Many themes available to customize the look of MobiLinc
- Quick device and scene control through ICONs

- Weather Support (Weather plug-in for ISY is required).



-SUDHANSHU KUMAR
ECE-E
Final year

ARTIFICIAL NEURAL NETWORK (ANN)

Introduction:

An Artificial Neural Network (ANN), usually called neural network (NN), is a mathematical model or computational model that is inspired by the structure and/or functional aspects of biological neural networks. A neural network consists of an interconnected group of artificial neurons, and it processes information using a connectionist approach to computation. In most cases an ANN is an adaptive system that changes its structure based on external or internal information that flows through the network during the learning phase. Modern neural networks are non-linear statistical data modeling tools. They are usually used to model complex relationships between inputs and outputs or to find patterns in data.

Neural networks versus conventional computers:

Neural networks take a different approach to problem solving than that of conventional computers.

Conventional computers use an algorithmic approach i.e. the computer follows a set of instructions in order to solve a problem. Unless the specific steps that the computer needs to follow are known the computer cannot solve the problem. That restricts the problem solving capability of conventional computers to problems that we already understand and know how to solve. But computers would be so much more useful if they could do things that we don't exactly know how to do.

Neural networks process information in a similar way the human brain does. The network is composed of a large number of highly interconnected processing elements (neurons) working in parallel to solve a specific problem. Neural networks learn by example. They cannot be programmed to perform a specific task. The examples must be selected carefully otherwise useful time is wasted or even worse the network might be functioning incorrectly. The disadvantage is that because the network finds out how to solve the problem by itself, its operation can be unpredictable.

On the other hand, conventional computers use a cognitive approach to problem solving; the way the problem is to solved must be known and stated in small unambiguous instructions. These instructions are then converted to a high level language program and then into machine code that the computer can understand. These machines are totally predictable; if anything goes wrong is due to a software or hardware fault.

Neural networks and conventional algorithmic computers are not in competition but complement each other. There are tasks are more suited to an algorithmic approach like arithmetic operations and tasks that are more suited to neural networks. Even more, a large number of tasks, require systems that use a



combination of the two approaches (normally a conventional computer is used to supervise the neural network) in order to perform at maximum efficiency.

Network layers:

The commonest type of artificial neural network consists of three groups, or layers, of units: a layer of "input" units is connected to a layer of "hidden" units, which is connected to a layer of "output" units.

The activity of the input units represents the raw information that is fed into the network.

The activity of each hidden unit is determined by the activities of the input units and the weights on the connections between the input and the hidden units.

The behavior of the output units depends on the activity of the hidden units and the weights between the hidden and output units.

This simple type of network is interesting because the hidden units are free to construct their own representations of the input. The weights between the input and hidden units determine when each hidden unit is active, and so by modifying these weights, a hidden unit can choose what it represents.

We also distinguish single-layer and multi-layer architectures. The single-layer organization, in which all units are connected to one another, constitutes the most general case and is of more potential computational power than hierarchically structured multi-layer organizations. In multi-layer networks, units are often numbered by layer, instead of following a global numbering.

Applications of neural networks Neural Networks in Practice:

Given this description of neural networks and how they work, what real world applications are they suited for? Neural networks have broad applicability to real world business problems. In fact, they have already been successfully applied in many industries.

Since neural networks are best at identifying patterns or trends in data, they are well suited for prediction or forecasting needs including:

- sales forecasting
- industrial process control
- customer research
- data validation
- risk management
- target marketing

But to give you some more specific examples; ANN are also used in the following specific paradigms: recognition of speakers in communications; diagnosis of hepatitis; recovery of telecommunications from faulty software; interpretation of multimeaning Chinese words; undersea mine detection; texture analysis; three-dimensional object recognition; hand-written word recognition; and facial recognition.

Neural networks in medicine:

Artificial Neural Networks (ANN) is currently a 'hot' research area in medicine and it is believed that they will receive extensive application to biomedical systems in the next few years. At the moment, the research is mostly on modeling parts of the human body and recognizing diseases from various scans

(e.g. cardiograms, CAT scans, ultrasonic scans, etc.).

Neural networks are ideal in recognizing diseases using scans since there is no need to provide a specific algorithm on how to identify the disease. Neural networks learn by example so the details of how to recognize the disease are not needed. What is needed is a set of examples that are representative of all the variations of the disease. The quantity of examples is not as important as the 'quality'. The examples need to be selected very carefully if the system is to perform reliably and efficiently.

Modeling and Diagnosing the Cardiovascular System:

Neural Networks are used experimentally to model the human cardiovascular system. Diagnosis can be achieved by building a model of the cardiovascular system of an individual and comparing it with the real time physiological measurements taken from the patient. If this routine is carried out regularly, potential harmful medical conditions can be detected at an early stage and thus make the process of combating the disease much easier.

A model of an individual's cardiovascular system must mimic the relationship among physiological variables (i.e., heart rate, systolic and diastolic blood pressures, and breathing rate) at different physical activity levels. If a model is adapted to an individual, then it becomes a model of the physical condition of that individual. The simulator will have to be able to adapt to the features of any individual without the supervision of an

expert. This calls for a neural network.

Another reason that justifies the use of ANN technology is the ability of ANNs to provide sensor fusion which is the combining of values from several different sensors. Sensor fusion enables the ANNs to learn complex relationships among the individual sensor values, which would otherwise be lost if the values were individually analyzed. In medical modeling and diagnosis, this implies that even though each sensor in a set may be sensitive only to a specific physiological variable, ANNs are capable of detecting complex medical conditions by fusing the data from the individual biomedical sensors.

Electronic noses:

ANNs are used experimentally to implement electronic noses. Electronic noses have several potential applications in telemedicine. Telemedicine is the practice of medicine over long distances via a communication link. The electronic nose would identify odors in the remote surgical environment. These identified odors' would then be electronically transmitted to another site where a door generation system would recreate them. Because the sense of smell can be an important sense to the surgeon, tele-smell would enhance telepresence surgery.

For more information on telemedicine and telepresence surgery click [here](#).

Instant Physician:

An application developed in the mid-1980s called the "instant physician" trained an auto associative memory neural network to store a large number of medical records, each of which includes information on symptoms, diagnosis, and treatment for a particular case. After training, the net can be presented with



input consisting of a set of symptoms; it will then find the full stored pattern that represents the "best" diagnosis and treatment.

Neural Networks in business:

Business is a diverted field with several general areas of specialization such as accounting or financial analysis. Almost any neural network application would fit into one business area or financial analysis.

There is some potential for using neural networks for business purposes, including resource allocation and scheduling. There is also a strong potential for using neural networks for database mining, which is, searching for patterns implicit within the explicitly stored information in databases. Most of the funded work in this area is classified as proprietary. Thus, it is not possible to report on the full extent of the work going on. Most work is applying neural networks, such as the Hopfield-Tank network for optimization and scheduling.

Marketing:

There is a marketing application which has been integrated with a neural network system. The Airline Marketing Tactician (a trademark abbreviated as AMT) is a computer system made of various intelligent technologies including expert systems. A feed forward neural network is integrated with the AMT and was trained using back-propagation to assist the marketing control of airline seat allocations. The adaptive neural approach was amenable to rule expression. Additionally, the application's environment changed rapidly and constantly, which required a continuously adaptive solution. The system is used to monitor and recommend booking advice

for each departure. Such information has a direct impact on the profitability of an airline and can provide a technological advantage for users of the system. [Hutchison & Stephens, 1987]

While it is significant that neural networks have been applied to this problem, it is also important to see that this intelligent technology can be integrated with expert systems and other approaches to make a functional system. Neural networks were used to discover the influence of undefined interactions by the various variables. While these interactions were not defined, they were used by the neural system to develop useful conclusions. It is also noteworthy to see that neural networks can influence the bottom line.

Credit Evaluation:

The HNC Company, founded by Robert Hecht-Nielsen, has developed several neural network applications. One of them is the Credit Scoring system which increases the profitability of the existing model up to 27%. The HNC neural systems were also applied to mortgage screening. A neural network automated mortgage insurance underwriting system was developed by the Nestor Company.

This system was trained with 5048 applications of which 2597 were certified. The data related to property and borrower qualifications. In a conservative mode the system agreed on the underwriters on 97% of the cases. In the liberal model the system agreed 84% of the cases. This is system run on an Apollo DN3000 and used 250K memory while processing a case file in approximately 1 sec.

Conclusion:

The computing world has a lot to gain from neural networks. Their ability to learn by example makes them very flexible and powerful. Furthermore there is no need to devise an algorithm in order to perform a specific task; i.e. there is no need to understand the internal mechanisms of that task. They are also very well suited for real time systems because of their fast response and computational times which are due to their parallel architecture.

Neural networks also contribute to other areas of research such as neurology and psychology. They are regularly used to model parts of living organisms and to investigate the internal mechanisms of the brain.

Perhaps the most exciting aspect of neural networks is the possibility that some day 'conscious' networks might be produced. There is a number of scientists arguing that consciousness is a 'mechanical' property and that 'conscious' neural networks are a realistic possibility.

Finally, I would like to state that even though neural networks have a huge potential we will only get the best of them when they are intergraded with computing, AI, fuzzy logic and related subjects.

-Vignesh Vicky
ECE –B
Prefinal year

DIGITAL PEN

The development of **digital pens** is comparable to the invention of email. This technology has just as much potential to revolutionize the way we share information and conduct business. Some businesses have been using the electronically-enabled pens for mapping,

surveying, filling out forms, and other business functions for the last few years, but consumers are just scratching the surface in understanding the multi-purpose functionality of these pens right now.



Some of the pens share technology components, while other components are unique. Some such as the Echo Smart pen, Pulse Smart pen and Capturx for OneNote require the use of special paper and contain a camera within the pen itself. For others such as the Mobile Digital Scribe, the recording/sensor technology is housed in an accompanying docking device that must be placed on or near the paper on which you are writing. All the pens we included in our review allow you to upload your hand-written words, tables and illustrations electronically to an application on your computer. Most convert the data to text which can then be imported into a Word document or the body of an email. Some provide online components to better help you share your electronic notes with friends and colleagues.

Computerized pens are very helpful for anyone who prefers to take hand-written notes over typing them into a laptop or PDA. Speaking of laptops, some bosses have banned them from meetings because of the tendency of those who bring them to play a more passive role. Without the ability to hide behind a computer screen, attendees are more likely to verbally share ideas and participate in the discussion. The electronic pens are less intrusive in a

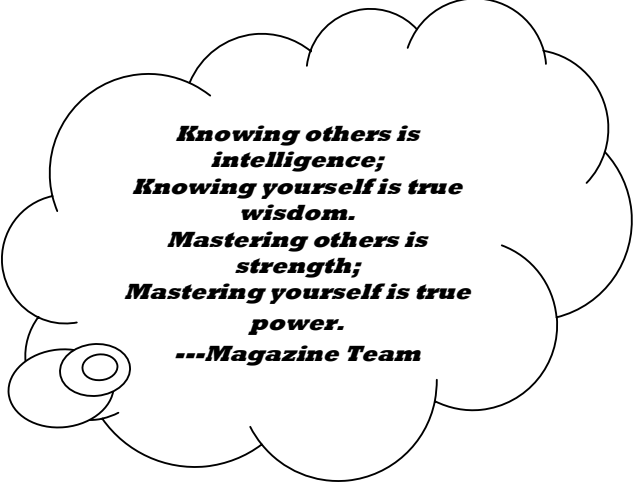


meeting, but they still allow meeting attendees to take notes which can then conveniently be imported to their computers for business use.

Email, instant messaging and texting capabilities today have caused daily, written communication to usually come in typed packages. Over the past few decades, we have seen a decreasing emphasis on perfecting handwriting skills in school as a result. Of course, as long as it's legible, handwriting will get the message across. Transmitting hand-written messages may seem unprofessional by some. However, there is a certain candor in handwriting that evokes friendship. The rarity of receiving a hand-written note actually draws attention to the message you wish to convey.

Digital pens will likely be a social revolution. Technology today provides for electronic writing devices that are comparable to the regular ink pens we are so familiar with. Handwriting in digital form allows for quick and convenient transmittal of messages to friends and colleagues.

- R.EUREKA SRI
ECE-A
Prefinal year



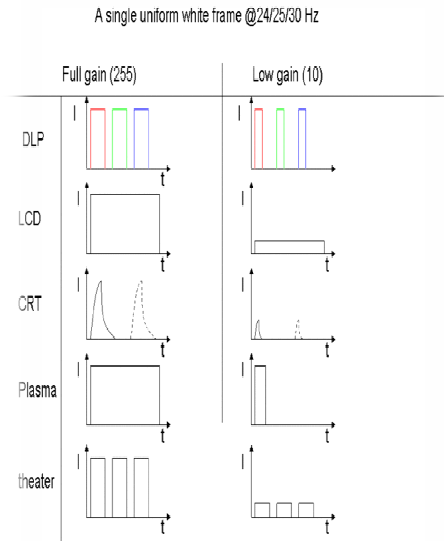
**Knowing others is
intelligence;
Knowing yourself is true
wisdom.
Mastering others is
strength;
Mastering yourself is true
power.
---Magazine Team**

DISPLAY TECHNOLOGY:

Different display technologies have vastly different temporal characteristics, leading to claimed perceptual differences for motion, flicker, etc.

The figure shows a sketch of how different technologies present a single white/grey frame. Time and intensity is not to scale. Notice that some have a fixed intensity, while the illuminated period is variable. This is a kind of pulse-width modulation. Others can vary the actual intensity in response to the input signal.

- Single-chip DLPs use a kind of "chromatic multiplexing" in which each color is presented serially. The intensity is varied by modulating the "on" time of each pixel within the time-span of one color. Multi-chip DLPs are not represented in this sketch, but would have a curve identical to the plasma display.
- LCDs have a constant (backlit) image, where the intensity is varied by blocking the light shining through the panel.
- CRTs use an electron beam, scanning the display, flashing a lit image. If interlacing is used, a single full-resolution image results in two "flashes". The physical properties of the phosphor are responsible for the rise and decay curves.
- Plasma displays modulate the "on" time of each sub-pixel, similar to DLP.
- Movie theaters use a mechanical shutter to illuminate the same frame 2 or 3 times, increasing the flicker frequency to make it less perceptible to the human eye.



-T.GAYATHRI
ECE –A
Prefinal year

ACTIVE POWER CONDITIONERS FOR HARMONICS

Business organizations are constantly looking for efficient energy solutions, especially with the government regulations getting stricter, and rising costs involved with power usage.

An active power conditioner which it claims will help in power factor (PF) correction, harmonics filtering, load balancing, neutral protection and bring a facility's energy consumption down by at least 50 per cent.

Icon is an insulated gate bipolar transistor IGBT-based active filter controlled by a digital signal processing (DSP) microcontroller. It senses the load requirement of a machine every 200 microseconds and thus manages to handle any fluctuation in the loads. It was developed by Shwetank Jain, an IITian,

with help from his professor Dr Gautam Poddar, as part of his BTech project. Later, he ventured out with his business organization P2 Power Solutions, an engineering solutions provider that focuses on energy-efficiency and power-quality enhancement.

Problems faced in normal scenario icon takes care of power quality issues like poor PF, harmonics in the supply system, unbalanced loads and over-current in the neutrals. "The normal electricity supply we get has a frequency of 50 hertz. However, due to some heavy loads and equipment like motors, drives, UPS, lighting loads and other non-linear loads, wave-forms of other frequencies which are odd multiples of 50, e.g., 150, 250, 350 or 550 hertz, are introduced in the supply system. As a result, harmonics currents cause over-heating of the cables and transformers, random tripping of circuit breakers or even damage to other machines connected to the system. Traditionally used capacitor-banks for PF improvement further intensify the ill-effects of the harmonic currents. This causes the maintenance costs, energy bills and other overhead costs to substantially rise up," illustrates Rajeev, head, and product development.

Plugging a power conditioner can help to produce a clean flow of regulate current to the equipment? Feels that icon is different from other power conditioners. "What sets icon apart from other power conditioners is its ability to sense the load requirement every 200 microseconds, and hence, manage severe fluctuations in the power distributed to equipment. Meanwhile, factor correction system (RTPFC) takes a response time of three to five milliseconds and capacitor-banks controlled by relays take up to three seconds to sense the load and then do the necessary correction. Therefore these traditional solutions are comparatively much slower in managing the power



fluctuations. Further, the ability of icon to handle the Holy Grails of power related issues—low PF, harmonics, unbalanced load and neutral current—sets it apart from all the available products in the market,” he says.

How it works?

The hardware and software are integrated through an assembly of correct rating, make and type of the various components, and a proper algorithm to control the working of these components. The two things interface through a central microcontroller installed in product. Icon comprises IGBTs, DC capacitors, microcontrollers, current sensors, voltage sensors, relays, inductor chokes and heat sinks. It also includes other common hardware components like power supplies, switchgear items, resistors, capacitors, cables/bus-bars and fans/blowers.

“The hardware of icon is quite similar to an inverter. We use self-made microcontrollers and the best components available in the market like IGBTs from Mitsubishi, Fujitsu or Infineon, and LEM sensors,”

icon uses P2 Power’s patent-applied software algorithm that is implemented through the two microcontrollers installed in the product. The software in the DSP microcontroller controls the functioning and operation of all the components assembled in the product. It identifies an effective physical layout and modifies the design of the load after observing various parameters, area of malfunction and the root cause of the problem.

“The microcontroller breaks down the current requirement of the load into harmonics, reactive current required to make the power factor unity, unbalanced current and neutral current. For example, if

100A reactive current, 50A unbalanced and 50A harmonics are required, a total of 200A is generated and is distributed to the loads as per the requirement. The software algorithm controls this distribution of current. The output current is controlled by IGBTs, which is also controlled by a microcontroller. We initially faced many problems in trying to stabilize the product and arrive at a perfect tuning between the product’s hardware and software. This resulted in several iterative improvements of the components and the software used to drive these components, thus, helping us make the product very stable and suitable in all electrical environments,” comments.

Application and cost:

After buying the product the user can connect icon anywhere in the supply system—to the transformer or to any desired sensitive electronic equipment. The power saved depends on the power factor of the equipment. “For example, if you maintain a high power factor of about 0.8, a minimum saving of 50 per cent is guaranteed. If your power factor is 0.4, you will save at least 75 per cent,

wherever there is a requirement of power, load-balancing harmonics is an important aspect. This product is suitable for any industry that consumes power and faces issues in load-balancing.

“It can be used in industries like cement, steel, paper, printing, hospitals and shopping malls, just to name a few. The government is trying to enforce regulations on the power consumption and the quality of power being maintained by the industries,”

icon has been priced significantly lower than similar products. As the product was developed in a university lab, costs on research and IP has been brought down. The product is available in various

current ratings including 15A, 35A, 70A, 100A, 140A, 210A, etc. These units are typically priced between Rs 3500 and Rs 4000 (excluding taxes) per ampere. However, these prices are not applicable to the units of rating lower than 70A.

Besides these costs the customer has to pay for the packaging and freight charges. P2 Power's scope of work covers the installation and commissioning of icon but the customer has to bear the basic cost of trucking and cabling of the unit, and installation of CTs at their site. All these are nominal costs and the total expenditure on installation never exceeds Rs 10,000. Apart from ensuring regular cleaning of the unit, and a dust- and moisture-free environment, icon doesn't require much maintenance. It's typical life is greater than 40,000 hours.

"The design of the system will keep on improving. Other special features based on the study of grid-condition at the client's sites and feedback from the client itself would guide the type of improvements to be incorporated in the system,"

-M.RAJA NIVASH
ECE-C
Second year



CLOUD COMPUTING

INTRODUCTION:

Cloud computing is the use of computing resources (hardware and software) that are delivered as a service over a network (typically the Internet). The name comes from the use of a cloud-shaped symbol as an abstraction for the complex infrastructure it contains in system diagrams. Cloud computing entrusts remote services with a user's data, software and computation. Cloud computing is at an early stage, with a motley crew of providers large and small delivering a slew of cloud-based services, from full-blown applications to storage services to spam filtering. Yes, utility-style infrastructure providers are part of the mix, but so are SaaS (software as a service) providers such as Salesforce.com. Today, for the most part, IT must plug into cloud-based services individually, but cloud computing aggregators and integrators are already emerging. InfoWorld talked to dozens of vendors, analysts, and IT customers to tease out the various components of cloud computing. Based on those discussions, here's a rough breakdown of what cloud computing is all about.

SAAS:

This types of cloud computing delivers a single application through the browser to thousands of customers using a multitenant architecture. On the customer side, it means no upfront investment in servers or software licensing; on the provider side, with just one app to maintain, costs are



low compared to conventional hosting. Salesforce.com is by far the best-known example among enterprise applications, but SaaS is also common for HR apps and has even worked its way up the food chain to ERP, with players such as Workday. And who could have predicted the sudden rise of SaaS "desktop" applications, such as Google Apps and Zoho Office?

Utility computing:

The idea is not new, but this form of cloud computing is getting new life from Amazon.com, Sun, IBM, and others who now offer storage and virtual servers that IT can access on demand. Early enterprise adopters mainly use utility computing for supplemental, non-mission-critical needs, but one day, they may replace parts of the datacenter. Other providers offer solutions that help IT create virtual datacenters from commodity servers, such as 3Tera's AppLogic and Cohesive Flexible Technologies' Elastic Server on Demand. Liquid Computing's LiquidQ offers similar capabilities, enabling IT to stitch together memory, I/O, storage, and computational capacity as a virtualized resource pool available over the network.

Web services in the cloud:

Closely related to SaaS, Web service providers offer APIs that enable developers to exploit functionality over the Internet, rather than delivering full-blown applications. They range from providers offering discrete business services -- such as Strike Iron and Xignite -- to the full range of APIs offered by Google Maps,

ADP payroll processing, the U.S. Postal Service, Bloomberg, and even conventional credit card processing services.

Platform as a service:

Another SaaS variation, this form of cloud computing delivers development environments as a service. You build your own applications that run on the provider's infrastructure and are delivered to your users via the Internet from the provider's servers. Like Legos, these services are constrained by the vendor's design and capabilities, so you don't get complete freedom, but you do get predictability and pre-integration. Prime examples include Salesforce.com's Force.com, Coghead and the new Google App Engine. For extremely lightweight development, cloud-based mashup platforms abound, such as Yahoo Pipes or Dapper.net.

MSP (managed service providers):

One of the oldest forms of cloud computing, a managed service is basically an application exposed to IT rather than to end-users, such as a virus scanning service for e-mail or an application monitoring service (which Mercury, among others, provides). Managed security services delivered by SecureWorks, IBM, and Verizon fall into this category, as do such cloud-based anti-spam services as Postini, recently acquired by Google. Other offerings include desktop management services, such as those offered by Center Beam or Ever dream.

Service commerce platforms:

A hybrid of SaaS and MSP, this cloud computing service offers a service hub that users interact with. They're most common in trading environments, such as expense management systems that allow users to order travel or secretarial services from a common platform that then coordinates the service delivery and pricing within the specifications set by the user. Think of it as an automated service bureau. Well-known examples include Rearden Commerce and Ariba.

Internet integration:

The integration of cloud-based services is in its early days. OpSource, which mainly concerns itself with serving SaaS providers, recently introduced the OpSource Services Bus, which employs in-the-cloud integration technology from a little startup called Boomi. SaaS provider Workday recently acquired another player in this space, Cape Clear, an ESB (enterprise service bus) provider that was edging toward b-to-b integration. Way ahead of its time, Grand Central -- which wanted to be a universal "bus in the cloud" to connect SaaS providers and provide integrated solutions to customers -- flamed out in 2005.

Today, with such cloud-based interconnection seldom in evidence, cloud computing might be more accurately described as "sky computing," with many isolated clouds of services which IT customers must plug into individually. On the other hand, as virtualization and SOA permeate the enterprise, the idea of loosely coupled services running on an agile, scalable infrastructure should eventually make every enterprise a node in the cloud.

It's a long-running trend with a far-out horizon. But among big megatrends, cloud computing is the hardest one to argue with in the long term.

Privacy:

The cloud model has been criticized by privacy advocates for the greater ease in which the companies hosting the cloud services control, thus, can monitor at will, lawfully or unlawfully, the communication and data stored between the user and the host company. Instances such as the secret NSA program, working with AT&T, and Verizon, which recorded over 10 million phone calls between American citizens, causes uncertainty among privacy advocates, and the greater powers it gives to telecommunication companies to monitor user activity. Using a cloud service provider (CSP) can complicate privacy of data because of the extent to which virtualization for cloud processing (virtual machines) and cloud storage are used to implement cloud service. The point is that CSP operations, customer or tenant data may not remain on the same system, or in the same data center or even within the same provider's cloud. This can lead to legal concerns over jurisdiction. While there have been efforts (such as US-EU Safe Harbor) to "harmonies" the legal environment, providers such as Amazon still cater to major markets (typically the United States and the European Union) by deploying local infrastructure and allowing customers to select "availability zones." Cloud computing poses privacy concerns because the service provider may access the data that is on the cloud at any point in time. They could accidentally or



deliberately alter or even delete information.

Postage and delivery services company, Pitney Bowes launched Volley, a cloud-based, digital mailbox service to leverage its communication management assets. They also faced the technical challenge of providing strong data security and privacy.

However, they were able to address the same concern by applying customized, application-level security, including encryption. Some of the companies researching cloud computing are big names in the computer industry. Microsoft, IBM and Google are investing millions of dollars into research. Some people think Apple might investigate the possibility of producing interface hardware for cloud computing systems.

Applications of cloud computing:

The applications of cloud computing are practically limitless. With the right middleware, a cloud computing system could execute all the programs a normal computer could run. Potentially, everything from generic word processing software to customized computer programs designed for a specific company could work on a cloud computing system.

Why would anyone want to rely on another computer system to run programs and store data? Here are just a few reasons:

Clients would be able to access their applications and data from anywhere at any time. They could access the cloud computing system using any computer

linked to the Internet. Data wouldn't be confined to a hard drive on one user's computer or even a corporation's internal network.

It could bring hardware costs down. Cloud computing systems would reduce the need for advanced hardware on the client side. You wouldn't need to buy the fastest computer with the most memory, because the cloud system would take care of those needs for you. Instead, you could buy an inexpensive computer terminal. The terminal could include a monitor, input devices like a keyboard and mouse and just enough processing power to run the middleware necessary to connect to the cloud system. You wouldn't need a large hard drive because you'd store all your information on a remote computer.

Corporations that rely on computers have to make sure they have the right software in place to achieve goals. Cloud computing systems give these organizations company-wide access to computer applications. The companies don't have to buy a set of software or software licenses for every employee. Instead, the company could pay a metered fee to a cloud computing company.

Servers and digital storage devices take up space. Some companies rent physical space to store servers and databases because they don't have it available on site. Cloud computing gives these companies the option of storing data on someone else's hardware, removing the need for physical space on the front end.

Corporations might save money on IT support. Streamlined hardware would, in theory, have fewer problems than a

network of heterogeneous machines and operating systems.

If the cloud computing system's back end is a grid computing system, then the client could take advantage of the entire network's processing power. Often, scientists and researchers work with calculations so complex that it would take years for individual computers to complete them. On a grid computing system, the client could send the calculation to the cloud for processing. The cloud system would tap into the processing power of all available computers on the back end, significantly speeding up the calculation.

-N.vignesh
ECE –B
Prefinal year

CURIOSITY-A MARS ROVER

“Curiosity is the passion that drives us through our everyday lives. We have become explorers and scientists with our need to ask questions and to wonder.”

Mars Science Laboratory (MSL) is a robotics pace mission to Mars launched by NASA on November 26, 2011, which successfully landed Curiosity, a Mars rover, in Gale Crater on August 6, 2012. The rover carries a variety of scientific instruments designed by an international team.

The Curiosity rover is a car-sized, rover. It was launched from Cape Canaveral on November 26, 2011 at 10:02 EST aboard the MSL spacecraft and successfully landed on Aeolis Palus in Gale Crater on Mars on August 6, 2012, 05:17 UTC. The Bradbury Landing site was less

than 2.4 km (1.5 mi) from the center of the rover's touchdown target after a 563,000,000 km (350,000,000 mi) journey.

The rover's goals include investigation of the Martian climate, geology, and whether Mars could have ever supported life, including investigation of the role of water and planetary habitability, as well as preparation for future human exploration.

The perfect image of CURIOSITY is



Details:

Operator:- NASA International team
Major contractors :- Boeing
Lockheed Martin
MacDonald Dettwiler

Mission type :- Rover

Launch date -November 26, 2011
15:02:00.211 UTC

Launch vehicle -Atlas V 541

Launch site -Cape Canaveral LC-41

Mission duration - 668 Martian sols (23 Earth months) primary mission.

COSPAR -ID2011-070A



Mass 900 kg (2,000 lb)-Mars landing
Date August 6, 2012, 05:17UTC SCET
MSD 49269 05:50 AMT
Coordinates Aeolis Pales
("Bradbury Landing") in Gale Crater.

Overview:

Curiosity is about twice as long and five times as heavy as the Spirit and Opportunity Mars exploration rover payloads of earlier U.S. Mars missions, and carries over ten times the mass of scientific instruments. MSL successfully carried out a more accurate landing than previous spacecraft to Mars, aiming for a small target landing ellipse of only 7 by 20 km (4.3 by 12 mi), in the Aeolis Pales region of Gale Crater. In the event, MSL achieved a landing only 2.4 kilometers (1.5 mi) from the center of the target. This location is near the mountain Aeolis Mons (a.k.a. "Mount Sharp"). The rover mission is set to explore for at least 687 Earth days (1 Martian year) over a range of 5 by 20 km (3.1 by 12 mi). NASA anticipates that the rover will function for at least the limit the parts were tested for, which is four years.

The Mars Science Laboratory mission is part of NASA's Mars Exploration Program, a long-term effort for the robotic exploration of Mars that is managed by the Jet Propulsion Laboratory of California Institute of Technology. The total cost of the MSL project is about US\$2.5 billion. Germany contributed 2.5 million Euros (\$3.1 million USD).

Previous successful U.S. Mars rovers include the Spirit and Opportunity,

and the Sojourner rover from the Mars Pathfinder mission.

Landing:

On August 6, 2012, Curiosity, the 900 kg robot technology, going anxious astronomers from NASA when attempting to land in the region Gale Crater, Mars. It must arise in the night between Sunday and Monday on the Red Planet, after a long journey of more than eight months.

"Mars is friendly to us, we will have good conditions for Sunday ... a dust storm spotted a few days ago has dissipated, giving way to a cloud of dust Benin enough,"

Said the scientist Ashwin Vasavada.

The \$ 2.5 billion invested in Curiosity will help answer a anguishing question, are we alone in the universe?

In fact the robot is in charge during the next 2 years, of a difficult task, find traces of past or present life on Mars. If there are traces of life on another planet in the solar system is that life is everywhere and is part of the matter in the universe. The landing of Curiosity is the most complex already made landings on Mars because the robot is too heavy for the impact is cushioned with balloons filled with air.

Risky 7 – Minutes:

To switch to 7 minutes, 21 243 km/h to 2.74 km/h, engineers had to design a huge "parachute" of 21 meters in diameter, equipped with retro-rockets that will allow robots to land like a feather. During those 7 minutes of terror, such as

NASA calls, three probes in orbit around Mars, grab all the signals sent by Curiosity for scientists to retransmit Pasadena.

Only after this high-risk approach that held its Curiosity Mars mission. With a nuclear generator, then leave it, eyes wide open for high-definition cameras, looking for an organic life. A laser will allow it to avoid obstacles that lie before him in closer than seven meters. Many other high-tech instruments to help the search for traces of life in a apparently sterile. Simple molecules of methane, enough scientists to happiness, as this gas are searched for the presence of life. To satisfy our curiosity, the robot will search, locate, drill, collect, analyze the matter of our universe and of course, send comments to us poor humans. But after more than eight months of "gestation" period of its journey into space, he will reach the risky step of its birth, of the Red Planet.

Steps risky landing Curiosity. Despite the countless images of Mars, the planet remains very mysterious. The red planet is the most visited by the probes, because it is closest to the Earth but also because its environment could resemble the Earth in the past. This is why scientists are creative and try whether this lost world could harbor life in the far or recent past. The Curiosity rover was designed to answer this question. Landing of Curiosity is scheduled on August 6 at 5:31 GMT.

Spacecraft:

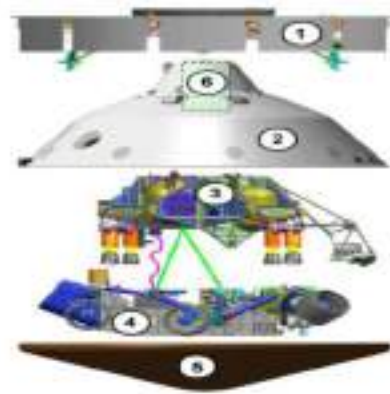
The spacecraft flight system had a mass at launch of 3,893 kg (8,580 lb), consisting of an Earth-Mars fuelled cruise stage (539 kg (1,190 lb)), the entry-descent-landing (EDL) system (2,401 kg (5,290 lb) + 390 kg (860 lb) of propellant),

and a 899 kg (1,980 lb) mobile rover with an integrated instrument package.

Goals and objectives:

The MSL mission has four scientific goals: Determine the landing site's habitability including the role of water, the study of the climate and the geology of Mars. It is also useful preparation for a future manned mission to Mars.

To contribute to these goals, MSL has eight main scientific objectives.



Biological:

Determine the nature and inventory of organic carbon compounds.

Inventory the chemical building blocks of life (carbon, hydrogen, nitrogen, oxygen, phosphorous, and sulfur).

Identify features that may represent the effects of biological processes (bio signatures).

Geological and geochemical:

Investigate the chemical, isotopic, and mineralogical composition of the Martian surface and near-surface geological materials.



Interpret the processes that have formed and modified rocks and soils.

Planetary process:

Assess long-timescale (i.e., 4-billion-year) Martian atmospheric evolution processes.

Determine present state, distribution, and cycling of water and carbon dioxide.

Surface radiation:

Characterize the broad spectrum of surface radiation, including galactic radiation, cosmic radiation, solar proton events and neutrons. As part of its exploration, it also measured the radiation exposure in the interior of the spacecraft as it traveled to Mars, and it is continuing radiation measurements as it explores the surface of Mars. This data would be important for a future manned mission.

- V.P.KALYAN KUMAR
ECE-B
Final year

BIO-MEDICAL INSTRUMENTATION

INTRODUCTION:

The bio-potential recorder is playing an important role in the biomedical instrumentation. Normally each doctor is performing his diagnosis based on the output from the recorder. If the recording is not faithful or the fidelity is very poor, then there is improper diagnosis. The recording instrumentation is developed in various stages.

During 1887, Waller recorded the first electrocardiogram using Callipary electrometer, immersion type electrodes and moving light sensitive paper. Then string galvanometer was introduced during 1903 by Einthoven, the father of Electrocardiography.

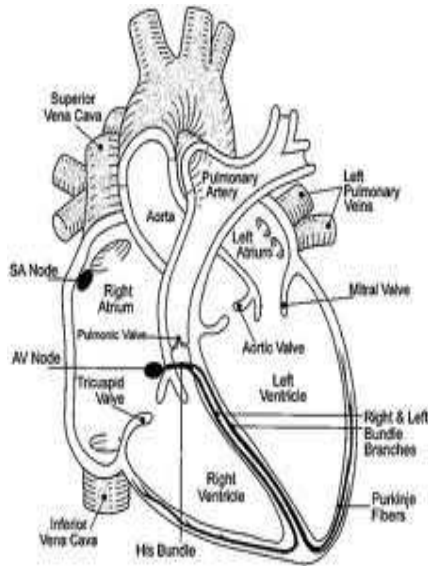
CHARACTERISTICS OF THE RECORDING SYSTEM:

The SENSITIVITY of the recording system is the magnitude of the input voltage required to produce a standard deflection in the record trace. Frequency response of the recorder is the frequency characteristics of the system such that variation of the sensitivity for different frequency components of the biosignal. Noise level increases the bandwidth of the system. It is therefore important to restrict the bandwidth to that required for faithful reproduction of the signal. A low noise level is maintained by special circuits and use of quality electronic components.

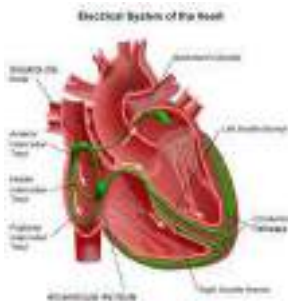
ECG:

ECG stands that Electrocardiography deals with the study of the electrical activity the HEART MUSCLES. The potential originated in the individual fibers of heart muscles are added to produce the ECG waveform. ECG sometimes called EKG which derived from German electrocardiogram. The electrocardiogram reflects the rhythmic electrical depolarization & repolarization of the myocardium (heart muscle) associated with the contractions of the atria and ventricles.

ANATOMY and CONDUCTING SYSTEM OF HEART:



CONDUCTING SYSTEM OF HEART:



ECG RECORDING SETUP:

The important parts of ECG recorder are listed as below:

- Patient cable & Defibrillator Protection Circuit
- Lead selector switch
- Calibrator
- Bio-amplifier

- Auxiliary Amplifier
- Isolated Power Supply
- Output Unit
- Power Switch

WORKING:

The ECG device detects and amplifies the tiny electrical changes on the skin that are caused when the heart muscle depolarizes during each heartbeat. At rest, each heart muscle cell has a negative charge, called the membrane potential, across its cell membrane. Decreasing this negative charge towards zero, via the influx of the positive cautions, is called depolarization, which activates the mechanisms in the cell that cause it to contract. During each heartbeat, a healthy heart will have an orderly progression of a wave of depolarization that is triggered by the cells in the senatorial node, spreads out through the atrium, passes through the atrioventricular node and then spreads all over the ventricles. This is detected as tiny rises and falls in the voltage between two electrodes placed either side of the heart which is displayed as a wavy line either on a screen or on paper. This display indicates the overall rhythm of the heart and weaknesses in different parts of the heart muscle.

Usually more than two electrodes are used, and they can be combined into a number of pairs (For example: left arm (LA), right arm (RA) and left leg (LL) electrodes form the three pairs LA+RA, LA+LL, and RA+LL). The output from each pair is known as a lead. Each lead looks at the heart from a different angle. Different types of EKGs can be referred to by the number of leads that are recorded, for example 3-lead, 5-lead or 12-lead



ECGs (sometimes simply "a 12-lead"). A 12-lead EKG is one in which 12 different electrical signals are recorded at approximately the same time and will often be used as a one-off recording of an ECG, traditionally printed out as a paper copy. 3- and 5-lead ECGs tend to be monitored continuously and viewed only on the screen of an appropriate monitoring device, for example during an operation or whilst being transported in an ambulance. There may or may not be any permanent record of a 3- or 5-lead ECG, depending on the equipment used.

This is the best way to measure and diagnose abnormal rhythms of the heart, particularly abnormal rhythms caused by damage to the conductive tissue that carries electrical signals, or abnormal rhythms caused by electrolyte imbalances. In a myocardial infarction (MI), the ECG can identify if the heart muscle has been damaged in specific areas, though not all areas of the heart are covered. The ECG cannot reliably measure the pumping ability of the heart, for which ultrasound-based (echocardiography) or nuclear medicine tests are used. It is possible for a human or other animal to be in cardiac arrest but still have a normal ECG signal (a condition known as pulse less electrical activity).

-REVATHY
ECE-E
second year

Wireless Charging Of Mobile Phones Using Micro Waves

With mobile phones becoming a basic part of life, the recharging of mobile phone batteries has always been a problem. The mobile phones vary in their talk time and battery standby according to their manufacturer and batteries. All these phones irrespective of their manufacturer and batteries have to be put to recharge after the battery has drained out. The main objective of their manufacturer and battery make. In this paper a new proposal has been made so as to make the recharging of the mobile phones is done automatically as you talk in your mobile phone. This is done by use of microwaves. The microwave signal is transmitted from the transmitter along with the message signal using special kind of antennas called slotted wave guide antenna at a frequency is 2.45GHz. There are minimal additions, which have to be made in the mobile handsets, which are the addition of a sensor, a "rectenna" and a filter. With the above setup, the need for separate chargers for mobile phones is eliminated and makes charging universal. Thus the more you talk, the mobile phones is eliminated and makes charging universal. Thus the more you talk, the more is your mobile phone charged! With this proposal the manufacturers would be able to remove the talk time and battery standby from their phone specifications.

INTRODUCTION:**THE ELECTROMAGNETIC SPECTRUM**

To start with, to know what a spectrum is: when white light is shone through a prism it is separated out into all the colours of the rainbow; this is the visible spectrum. So white light is a mixture of all colors. Black is NOT a color, it is what you get when all the light is taken away. Some physicists pretend that light consists of tiny particles which they call photons. They travel at the speed of light (what a surprise). The speed of light is about 300,000,000 meters per second. When they hit something they might bounce off, go right through or get absorbed. What happens depends a bit on how much energy they have. If they bounce off something and then go into your eye you will "see" the thing they have bounced off. Some things like glass and Perspex will let them go through; these materials are transparent. Black objects absorb the photons so you should not be able to see black things, you will have to think about this one. These poor old physicists get a little bit confused when they try to explain why some photons go through a leaf, some are reflected, and some are absorbed. They say that it is because they have different amounts of energy.

Other physicists pretend that light is made of waves. These physicists measure the length of the waves and this helps them to explain what happens when light hits leaves. The light with the longest wavelength (red) is absorbed by the green stuff (chlorophyll) in the leaves. So is the light with the shortest wavelength (blue). In between these two colors there is green

light, this is allowed to pass right through or is reflected. (Indigo and violet have shorter wavelengths than blue light.) Well it is easy to explain some of the properties of light by pretending that it is made of tiny particles called photons and it is easy to explain other properties of light by pretending that it is some kind of wave.

The visible spectrum is just one small part of the electromagnetic spectrum. These electromagnetic waves are made up of two parts. The first part is an electric field. The second part is a magnetic field. So that is why they are called electromagnetic waves. The two fields are at right angles to each other.

THE MICROWAVE REGION

Microwave wavelengths range from approximately one millimeter (the thickness of a pencil lead) to thirty centimeters (about twelve inches). In a microwave oven, the radio waves generated are tuned to frequencies that can be absorbed by the food. The food absorbs the energy and gets warmer. The dish holding the food doesn't absorb a significant amount of energy and stays much cooler. Microwaves are emitted from the Earth, from objects such as cars and planes, and from the atmosphere. These microwaves can be detected to give information, such as the temperature of the object that emitted the microwaves.

Microwaves have wavelengths that can be measured in centimeters! The longer microwaves, those closer to a foot in length, are the waves which heat our food in a microwave oven. Microwaves



are good for transmitting information from one place to another because microwave energy can penetrate haze, light rain and snow, clouds, and smoke. Shorter microwaves are used in remote sensing. These microwaves are used for radar like the Doppler radar used in weather forecasts. Microwaves, used for radar, are just a few inches long. Because microwaves can penetrate haze, light rain and snow, clouds and smoke, these waves are good for viewing the Earth from space. Microwave waves are used in the communication industry and in the kitchen as a way to cook foods. Microwave radiation is still associated with energy levels that are usually considered harmless except for people with pace makers. Here we are going to use the S band of the Microwave Spectrum. Microwave frequency bands

Designation	Frequency range
L Band	1 to 02 GHz
S Band	2 to 04 GHz
C Band	4 to 08 GHz
X Band	8 to 12 GHz
Ku Band	12 to 18 GHz
K Band	18 to 26 GHz
Ka Band	26 to 40 GHz
Q Band	30 to 50 GHz
U Band	40 to 60 GHz
V Band	46 to 56 GHz
W Band	56 to 100 GHz

The frequency selection is another important aspect in transmission. Here we have selected the license free 2.45 GHz ISM band for our purpose. The Industrial, Scientific and Medical (ISM) radio bands were originally reserved internationally for non-commercial use of RF

electromagnetic fields for industrial, scientific and medical purposes. The ISM bands are defined by the ITU-T in S5.138 and S5.150 of the Radio Regulations. Individual countries use of the bands designated in these sections may differ due to variations in national radio regulations. In recent years they have also been used for license-free error-tolerant communications applications such as wireless LANs and Bluetooth, 900 MHz band (33.3 cm) (also GSM communication in India) 2.45 GHz band (12.2 cm) IEEE 802.11b wireless Ethernet also operates on the 2.45 GHz band

TRANSMITTER DESIGN:

The MAGNETRON (A) is a self-contained microwave oscillator that operates differently from the linear-beam tubes, such as the TWT and the klystron. View (B) is a simplified drawing of the magnetron. CROSSED-ELECTRON and MAGNETIC fields are used in the magnetron to produce the high-power output required in radar and communications equipment.

The magnetron is classed as a diode because it has no grid. A magnetic field located in the space between the plate (anode) and the cathode serves as a grid. The plate of a magnetron does not have the same physical appearance as the plate of an ordinary electron tube. Since conventional inductive-capacitive (LC) networks become impractical at microwave frequencies, the plate is fabricated into a cylindrical copper block containing resonant cavities that serve as tuned circuits. The magnetron base differs

considerably from the conventional tube base. The magnetron base is short in length and has large diameter leads that are carefully sealed into the tube and shielded. The cathode and filament are at the center of the tube and are supported by the filament leads. The filament leads are large and rigid enough to keep the cathode and filament structure fixed in position. The output lead is usually a probe or loop extending into one of the tuned cavities and coupled into a waveguide or coaxial line. The plate structure, shown in figure 2 -18, is a solid block of copper. The cylindrical holes around its circumference are resonant cavities. A narrow slot runs from each cavity into the central portion of the tube dividing the inner structure into as many segments as there are cavities. Alternate segments are strapped together to put the cavities in parallel with regard to the output. The cavities control the output frequency. The straps are circular, metal bands that are placed across the top of the block at the entrance slots to the cavities. Since the cathode must operate at high power, it must be fairly large and must also be able to withstand high operating temperatures. It must also have good emission characteristics, particularly under return bombardment by the electrons. This is because most of the output power is provided by the large number of electrons that are emitted when high-velocity electrons return to strike the cathode. The cathode is indirectly heated and is constructed of a high-emission material. The open space between the plate and the cathode is called the INTERACTION SPACE. In this space the electric and magnetic fields interact to exert force upon the electrons.

RECEIVER DESIGN:

The basic addition to the mobile phone is going to be the rectenna. A rectenna is a rectifying antenna, a special type of antenna that is used to directly convert microwave energy into DC electricity. Its elements are usually arranged in a mesh pattern, giving it a distinct appearance from most antennae. A simple rectenna can be constructed from a schottky diode placed between antenna dipoles. The diode rectifies the current induced in the antenna by the microwaves. Rectennae are highly efficient at converting microwave energy to electricity. In laboratory environments, efficiencies above 90% have been observed with regularity. Some experimentation has been done with inverse rectennae, converting electricity into microwave energy, but efficiencies are much lower--only in the area of 1%. With the advent of nanotechnology and MEMS the size of these devices can be brought down to molecular level. It has been theorized that similar devices, scaled down to the proportions used in nanotechnology, could be used to convert light into electricity at much greater efficiencies than what is currently possible with solar cells. This type of device is called an optical rectenna. Theoretically, high efficiencies can be maintained as the device shrinks, but experiments funded by the United States National Renewable Energy Laboratory have so far only obtained roughly 1% efficiency while using infrared light. Another important part of our receiver circuitry is a simple sensor. This is simply used to identify when the mobile phone user is talking. As our main objective is to charge the mobile phone with the transmitted microwave



after rectifying it by the rectenna, the sensor plays an important role.

THE PROCESS OF THE RECTIFICATION:

Studies on various microwave power rectifier configurations show that a bridge configuration is better than a single diode one. But the dimensions and the cost of that kind of solution do not meet our objective. This study consists in designing and simulating a single diode power rectifier “in hybrid technology” with improved sensitivity at low power levels. We achieved good matching between simulation results and measurements thanks to the optimization of the packaging of the Schottky diode.

Microwave energy transmitted from space to earth apparently has the potential to provide environmentally clean electric power on a very large scale. The key to improve transmission efficiency is the rectifying circuit. The aim of this study is to make a low cost power rectifier for low and high power levels at a frequency of 2.45 GHz with good efficiency of rectifying operation. The objective also is to increase the detection sensitivity at low levels of power. Different configurations can be used to convert the electromagnetic wave into DC signal, the study done in showed that the use of a bridge is better than a single diode, but the purpose of this study is to achieve a low cost microwave rectifier with single Schottky diode for low and high power levels that has a good performances. This study is divided on two kinds of technologies the first is the hybrid technology and the second is the monolithic one. The goal of this

investigation is the development of a hybrid microwave rectifier with single Schottky diode. The first study of this circuit is based on the optimization of the rectifier in order to have a good matching of the input impedance at the desired frequency 2.45GHz. Besides, the aim of the second study is the increasing of the detection sensitivity at low levels of power.

SENSOR CIRCUITRY:

The sensor circuitry is a simple circuit, which detects if the mobile phone receives any message signal. This is required, as the phone has to be charged as long as the user is talking. Thus a simple F to V converter would serve our purpose. In India the operating frequency of the mobile phone operators is generally 900MHz or 1800MHz for the GSM system for mobile communication. Thus the usage of simple F to V converters would act as switches to trigger the rectenna circuit to on.

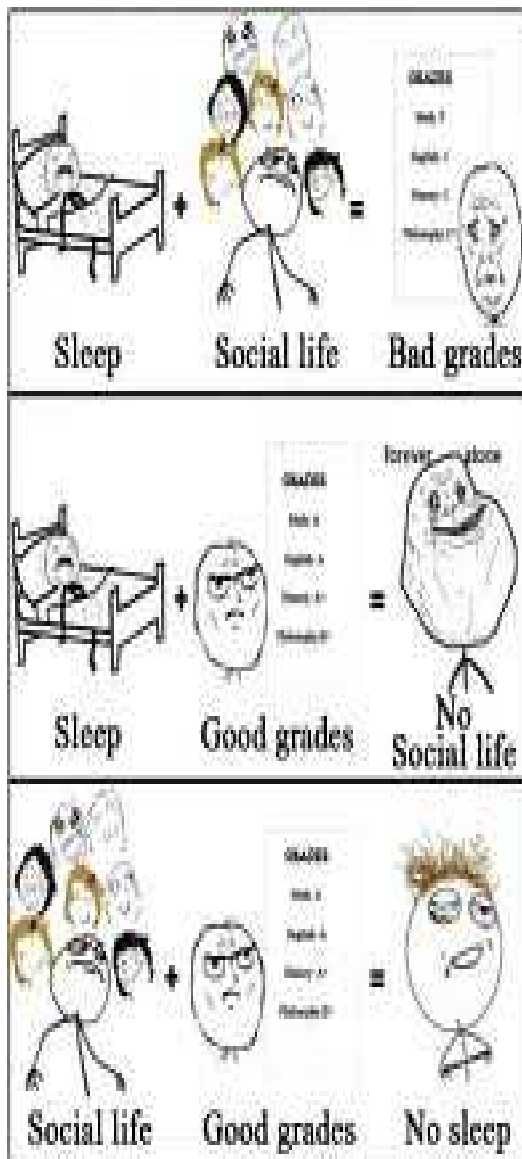
A simple yet powerful F to V converter is LM2907. Using LM2907 would greatly serve our purpose. It acts as a switch for triggering the rectenna circuitry. The general block diagram for the LM2907 is given below. Thus, on the reception of the signal the sensor circuitry directs the rectenna circuit to ON and the mobile phone begins to charge using the microwave power.

CONCLUSION

Thus this paper successfully demonstrates a novel method of using the power of the microwave to charge the mobile phones without the use of wired chargers. Thus this method provides great advantage to

the mobile phone users to carry their phones anywhere even if the place is devoid of facilities for charging. A novel use of the rectenna and a sensor in a mobile phone could provide a new dimension in the revelation of mobile phone.

- B.NagaNandhini
ECE-B
Third Year



Values to Imbibe:

The most **selfish** letter,

I----Avoid it.

The most **satisfying** two lettered word,

We--- Use it.

The most **damaging** three lettered word

Ego-- Quit it.

The most **pleasing** four lettered word

Love -- Value it.

The most **friendly** five lettered word

Smile-- Keep it.

The most **spreading** six lettered word,

Rumor-- Ignore it.

The most **admirable** seven lettered word,

Success--Achieve it.

The most **harmful** eight lettered word,

Jealousy-- Distance it.

The most **powerful** nine lettered word,

Knowledge--Acquire it.

The most **supportive** ten lettered word,

Confidence-- Build it.

--- **Maaazine Team**



NON-TECHNICAL ARTICLES

“ENJOY THE PROCESS OF YOUR LIFE”

Peter was a very lively little boy. Everyone loved him. But he had one weakness unlike other humans. What was that?

Peter could never live and enjoy the moment at present. He failed to enjoy the process of life. When he was in school, he dreamed of playing in home with his friends. When he was in home, he dreamed of his annual examinations.

One morning, Peter was out walking in a park near his home. Feeling tired, he decided to take rest. After few minutes of sleep, he heard someone calling his name. “Peter!” As he slowly opened his eyes, he was shocked by seeing a women standing above him. She must have been over a 80 years old. In the women’s hand was a magical little ball with hole at centre and golden thread.

‘Peter’ she said. ‘This is the thread of your life’. If you pull a thread just a bit, an hour will pass in seconds. If yo pull a little harder, whole day will pass in minutes.

And if you pull the thread with all your strength even years will pass by in days. Peter was very much excited and eager to use the magical ball.

The next day, Peter was sitting in classroom feeling tired and bored. Suddenly he remembered the ball. As he pulled a little nit, he found himself at home playing with his friends realizing the power of ball and pulled the thread again. Suddenly he turned into a teen ager with young girl friend. He was also got bored by his adolescent age and he pulled the thread again.

Now he found that he had transformed into a adult and got married with his girlfriend. At present he had 3 kids.

Again he pulled the thread and turned to a ninety year old man with white hair and his wife was passed away. His children were grown up and got married.

Peter realized that he had not taken time to enjoy the moment.

Peter was sad at this juncture and he decided to go out to park where he used to walk as a boy to clear his worries.

As he entered the park, suddenly he heard a same voice from the lawn. 'Peter! Peter! H E LOOKED UP TO SEE THAT IT WAS NONE OTHER THAN OLD WOMAN. She asked peter that "How have you enjoyed my special gift?

Peter replied that at first I was very happy and having fun but now I hate it. My whole life was passed within couple of days. There could have been sad times as well as great times but I didn't experience I feel guilty. The old woman replied that she could give him another chance. Peter was shocked and accepted the chance. He asked the women what she is going to give for him. She told that "you can ask me anything without fail I will do it for".

Peter thought for an instant and asked the woman to give back his childhood days.

Here comes a twist for you. Again he heard someone calling his name and opened his eyes 'who could be this time? When he opened his eyes, he was happy to see his mother standing over him. She looked young, healthy. Later Peter realized that he had dreamt of being an old man.

'Hurry up my boy'. It is getting late. Get ready to go to school. Peer was happy and he ran to get ready for studies. We here in real world will

never get a second chance to live life. Today is your chance to awaken to gift of living. Let this new day be the defining moment of our life, the day that we make decision once and for all to focus is really important. Do the things that you have always wanted to do . Stop putting off your happiness for sake of achievement. Instead, why not enjoy the process?

Start tending to your soul.

--R.KARTHICK GOWTHAM
ECE-B
Final year

MEANING FOR A GOOD STUDENT

- A - Active**
- G - Goes to collage regularly**
- O - Obey parents & staffs**
- O - Orderness in thinking**
- D - Does assignments in time**
- S - Serves the needy**
- T - Takes the right direction**
- U - Uses the books in right way**
- D - Does not waste time**
- E - Engages in useful activities**
- N - Never hurts others**
- T - Take advice from others**

I.BENCIA
ECE-A
PREFINAL YEAR



BRAIN TEASERS

1. QUESTION:

There is a safe with a 5 digit number as the key. The 4th digit is 4 greater than the second digit, while the 3rd digit is 3 less than the 2nd digit. The 1st digit is thrice the last digit. There are 3 pairs whose sum is 11. Find the number.

ANSWER:

65292 As per given conditions, there are three possible combinations for 2nd, 3rd and 4th digits. They are (3, 0, 7) or (4, 1, 8) or (5, 2, 9) It is given that there are 3 pairs whose sum is 11. All possible pairs are (2, 9), (3, 8), (4, 7), (5, 6). Now required number is 5 digit number and it contains 3 pairs of 11. So it must not be having 0 and 1 in it. Hence, the only possible combination for 2nd, 3rd and 4th digits is (5, 2, 9) Also, 1st digit is thrice the last digit. The possible combinations are (3, 1), (6, 2) and (9, 3), out of which only (6, 2) with (5, 2, 9) gives 3 pairs of 11. Hence, the answer is 65292.

2. QUESTION:

A person travels on a cycle from home to church on a straight road with wind against him. He took 4 hours to reach there. On the way back to the home, he took 3 hours to reach as wind was in the same direction. If there is no wind, how much time does he take to travel from home to church?

ANSWER:

Let distance between home and church is D. A person took 4 hours to reach church. So speed while travelling towards church is $D/4$. Similarly, he took 3 hours to reach home. So speed while coming back is $D/3$. There is a speed difference of $7*D/12$, which is the wind helping person in 1 direction, & slowing him in the other direction. Average the 2 speeds, & you have the speed that person can travel in no wind, which is $7*D/24$. Hence, person will take $D / (7*D/24)$ hours to travel distance D which is $24/7$ hours. Answer is 3 hours 25 minutes 42 seconds

3. QUESTION:

If a bear eats 65 pounds of fish every day EXCEPT every 6th day which it only eats 45 pounds of fish. If the bear continues this, how many pounds of fish will it eat in 200 days?

ANSWER:

The bear will eat 12,340 pounds of fish in 200 days. It is given that on every 6th day bear eats 45 pounds of fish i.e. on day number 6, 12, 18, 24, 192, 198 the bear eats 45 pounds of fish. Total number of 6th days = $200/6 = 33$ (the bear eats 45 pounds) Hence, the normal days are = $200 - 33 = 167$ (the bear eats 65 pounds) Thus, in 200 days, the bear will eat = $(167) * (65) + (33) * (45) = 10855 + 1485 = 12,340$ pounds

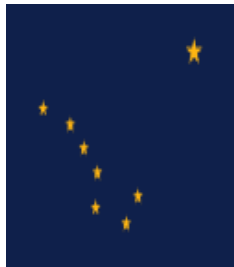
-VANISHREE.A

ECE-E

Final year

ALASKA

Alaska is the largest state in the United States by area. It is situated in the northwest extremity of the North American continent, with Canada to the east, the Arctic Ocean to the north, and the Pacific Ocean to the west and south, with Russia further west across the Bering Strait. Alaska is the 4th least populous and the least densely populated of the 50 United States. Approximately half of Alaska's 722,718 residents live within the Anchorage metropolitan area.



FLAG



SEAL

Capital

Juneau

Density

1.26/sq mi (0.49/km²)
Ranked 50th in the U.S.

Median income**household**

US\$64,333 (4th)

Largest city

Anchorage

Admission to Union

January 3, 1959 (49th)

Governor

Sean Parnell

Lieutenant Governor

Mead Treadwell

**GEOGRAPHY:****Area**

Ranked 1st in the U.S.

Total

663,268 sq m
(1,717,854 km²)

% water

13.77

Latitude

51°20'N to 71°50'N

Longitude

130°W to 172°E

Population

Ranked 47th in the
U.S.

Mount McKinley is both the highest peak
Augustine Volcano erupting on
in
Alaska and in all of North America.



the Augustine Volcano erupting on
January 12, 2006.



Climate:

The climate in Juneau and the southeast panhandle is a mid-latitude oceanic climate classification in the southern sections and a subarctic oceanic climate in the northern parts. On an annual basis, the panhandle is both the wettest and warmest part of Alaska with milder temperatures in the winter and high precipitation throughout the year. This is also the only region in Alaska in which the average daytime high temperature is above freezing during the winter months.

The area gets less rain than southeast Alaska, it gets more snow, and days tend to be clearer. On average, Anchorage receives 16 inches (406 mm) of precipitation a year, with around 75 inches (191 cm) of snow, although there are areas in the south central which receive far more snow. It is a subarctic climate due to its brief, cool summers.



Barrow, known colloquially for many years by the nickname "Top of the

World", is the northernmost city in the United States.

The climate of the interior of Alaska is subarctic. Some of the highest and lowest temperatures in Alaska occur around the area near Fairbanks. The summers may have temperatures reaching into the 90s °F (the low to mid 30s °C), while in the winter, the temperature can fall below -60 °F (-51.1 °C). Precipitation is sparse in the Interior, often less than 10 inches (254 mm) a year, but what precipitation falls in the winter tends to stay the entire winter.

The highest is 100 °F (37.8 °C) in Fort Yukon on June 27, 1915 making Alaska tied with Hawaii as the state with the lowest high temperature in the United States. The lowest official Alaska temperature is -80 °F (-62.2 °C) in Prospect Creek on January 23, 1971, one degree above the lowest temperature recorded in continental North America (in Snag, Yukon, Canada).

The climate in the extreme north of Alaska is Arctic (KöppenET) with long, very cold winters and short, cool summers. Even in July, the average low temperature in Barrow is 34 °F (1.1 °C). Precipitation is light in this part of Alaska, with many places averaging less than 10 inches (254 mm) per year, mostly as snow which stays on the ground almost the entire year.

ECONOMY RELIGION



Oilfield facilities St. Michael's Russian Orthodox Cathedral in Sitka

The 2007 gross state product was \$44.9 billion, 45th in the nation. Its per capita personal income for 2007 was \$40,042, ranking 15th in the nation. The oil and gas industry dominates the Alaskan economy, with more than 80% of the state's revenues derived from petroleum extraction. Alaska's main export product (excluding oil and natural gas) is seafood, primarily salmon, cod, Pollock and crab. Agriculture represents only a fraction of the Alaskan economy. Agricultural production is primarily for consumption within the state and includes nursery stock, dairy products, vegetables, and livestock. Manufacturing is limited, with most foodstuffs and general goods imported from elsewhere. Employment is primarily in government and industries such as natural resource extraction, shipping, and transportation. Military bases are significant component of the economy in both Fairbanks and Anchorage. Tourists have contributed to the economy by supporting local lodging.

The Trans-Alaska Pipeline transports oil, Alaska's most

financially important export, from the North Slope to Valdez. Pertinent are the heat pipes in the column mounts, which disperses heat upwards and prevents melting of permafrost.

Alaska's economy depends heavily on increasingly expensive diesel fuel for heating, transportation, electric power and light. Though wind and hydroelectric power are abundant and underdeveloped, proposals for state-wide energy systems were judged uneconomical due to low fuel prices, long distances and low population.

The cost of a US gallon of gas in urban Alaska today is usually \$0.30–\$0.60 higher than the national average; prices in rural areas are generally significantly higher but vary widely depending on transportation costs, seasonal usage peaks, nearby petroleum development infrastructure and many other factors.

Alaska accounts for one-fifth (20 percent) of domestically produced United States oil production. Prudhoe Bay (North America's largest oil field) alone accounts for 8% of the U.S. domestic oil production.



BP's Alaska headquarters in Anchorage.



Alaska Airlines Boeing 737 taking off from Ted Stevens Anchorage International Airport.



Alyeska Prince Hotel in Girdwood.



The Kachemak Bay campus of the University of Alaska Anchorage.

STATE SYMBOLS:

The "Forget-me-not" is the state's official flower



- State Motto: North to the Future
- Nicknames: "The Last Frontier" or "Land of the Midnight Sun" or "Seward's Icebox"
- State bird: Willow Ptarmigan.
- State fish: King Salmon.
- State fossil: Woolly Mammoth.
- State insect: dragon-fly.
- State land mammal: Moose.
- State marine mammal: Bowhead Whale.
- State mineral: Gold.
- State song: "Alaska's Flag"
- State sport: Dog Mushing.
- State tree: Sitka Spruce.
- State dog: Alaskan Malamute.
- State soil: Tanana.



Alaska's popular annual events are the
Bethel is the largest city in rural



Iditarod Trail Sled Dog Race. Alaska.

-P.FathimaSanofer
ECE B
Final year

How to Stay Positive

The ability to stay positive is one of the most important skills that you can develop. Why is this? Everyone wants to know deep down inside that they were put here to do something great. The problem is that life is tough. Most people let these challenges rob them from the life they really want. Most people will never be able to realize their dreams.



Sometimes the truth can sting a bit but it's also necessary to know. If you want to achieve greatness, if you want to experience the best life possible, then be prepared to get knocked down over and over. Staying positive during situations like this, is not easy. This explains why there are so many people who have stopped trying to reach their dreams.

So how do you remain positive regardless of what type of challenge you are facing? It all comes down to how you choose to respond to any situation as well as what you decide to focus on. No matter how bad your situation is, you can always find something good in it. So what you can do is start focusing on what is good. Even if you lost your job and you're

about to be evicted from your home, you can still be positive.

Knowing how to be positive can help you overcome a lot of challenges that causes many others to fail. So what can you do? You can do things like read positive books or listen to songs with positive lyrics.

-R.BADRINATH
ECE - E
Second year

Learn to Relax

Being able to learn to relax can be very important especially if you have a pretty stressful life. When you build up too much tension without being able to control or release it,



overtime, you can develop some serious health problems. This can be both mental and physical.

If you want to learn how to relax, then what you may want to consider doing is study the people who seem to be the most at ease. So who are these people? The answer is children. If you compare the stress levels of a working adult and a child, there is no comparison. The child has way less stress than the adult and because of this, it's easier for them to relax. You may say that's because they don't have anything to worry about, there are no demands, and no responsibilities to take care of. That is true



Knowing how to relax can be as simple as keeping things simple. When we worry about things and constantly focus on what we need to do, it's hard to relax. So what can you do? One way is to realize that most of the things that we worry about never even happen. It's good to think ahead and cover your bases to make sure that future problems don't show up but many people spend a lot of time worrying about things that are totally out of their control. Learn to relax by only

focusing on things you can control. Relaxing doesn't always mean to sit back and not do anything. Again; it comes down to your focus.

Children are able to relax because their focus on things that are in front of them and not on things in the future that may not even come true. When you are not working, you can do things like meditate or even get a massage. The key is to simplify your life. You can still relax and do all the things you need to do

-R.BADRINATH
ECE - E
Second year

Top 5 Tips To Make Your Computer Work Faster

Computer technology keeps on evolving everyday and during this evolution new things easily get old and are replaced almost every day. Faster machines keep coming in the market to flush out the old ones. And I know faster things are really attractive be it anything, bike or computer. So everyone who's got a old, rather slower computer wants to increase the efficiency of his/her computer and at the same time wants to make it faster. So, here we have got the list of Top 5 Tips To Make Your Computer Faster.

Don't Do Multitasking

I have seen people who simultaneously play games, listen to music, surf the Internet, view photos

etc and later complain that their computer isn't fast enough. Well let me tell you that you should avoid doing this thing because it occupies more space on your RAM and thereby making your computer work slowly. So, I advise you to make the priority list and things which are of more priority than the ones of less priority.

Avoid Overload

Another way of making your computer faster. What people do nowadays is that they install all kinds of softwares in their computer, even those which they don't need and this makes their system heavy when you run the softwares you need a lot. Overloading should be avoided, computer gets overloaded when you run quite a lot heavy programs simultaneously.

Increase RAM

RAM stands for Random Access Memory and this is the memory which provides space for the things you do on your computer. The more RAM you will have on your computer, the more memory you will get and the more memory will lead to greater efficiency and faster speed. RAM can be purchased from market, there are different versions of RAM like DDR1 DDR2 DDR3 etc and the less available it is in the market, the more expensive it is subjected to be.

Choose OS Wisely

Operating system is everything for a computer. Every OS is basically designed to be fast but still the better UI the OS will have, the heavier it will be. So choose your OS wisely. Let me give you an example, Linux is amazingly fast than Windows because Linux is command based and less user friendly while Windows is more user friendly and has heavy GUI.

Stay Light

Always try to keep your computer as light as possible. Don't install too much software, don't store too much data. Always try to keep 15% of your total storage's size empty. Staying light always help. That's why your computer is fast when it's newer than when it's old because a new computer doesn't have a lot of data stored in it.

-SUDHANSHUKUMAR

ECE E

Final year

Interesting Facts About English

1. The most common letter in English is "e".
2. The most common vowel in English is "e", followed by "a".
3. The most common consonant in English is "r", followed by "t".
4. Every syllable in English must have a vowel (sound). Not all syllables have consonants.



5. Only two English words in current use end in "-gry". They are "angry" and "hungry".
6. The word "bookkeeper" (along with its associate "bookkeeping") is the only unhyphenated English word with three consecutive double letters. Other such words, like "sweet-toothed", require a hyphen to be readily readable.
7. The word "triskaidekaphobia" means "fear of Friday the 13th". It also means "superstition about the number thirteen" in general.
8. More English words begin with the letter "s" than with any other letter.
9. A preposition is always followed by a noun (i.e. noun, proper noun, pronoun, noun group, gerund).
10. The word "uncopyrightable" is the longest English word in normal use that contains no letter more than once.
11. A sentence that contains all 26 letters of the alphabet is called a "pangram".
12. The following sentence contains all 26 letters of the alphabet: "The quick brown fox jumps over the lazy dog." This sentence is often used to test typewriters or keyboards. –

H.GAAYATHRI
ECE-B
Final year

Cricket in Examination

1. Preparation for Examination--- **Fielding**
2. Tough Question Paper-----**Fast Bowling**
3. Easy Question Paper----- **Slow Bowling**
4. Confusing Question Paper---- **Spin Bowling**
5. Unexpected Question Paper—**Googly**
6. Picked while cheating----- **Catch out**
7. First class with honours-----**Sixer**
8. First class-----**Four runs**
9. Just pass -----**One run**
10. Caught while helping others--- **Hit wicket**

--Magazine Team

Winners & losers

Winners	Losers
• The winners always the part of answers.	• The loser is always the part of problems.
• The winners says, "It may be difficult but is possible".	• The loser says, "It may be possible but is too difficult".
• Winners are a part of the team.	• Losers are apart from the team.
• Winners see the gain.	• Losers see the pain.
• Winners are like thermostat.	• Losers are like thermometers

**-I.BENCIA
ECE-A
PREFINAL YEAR**

Researchers state only 40 Years of Oil and 65 Years of Gas left

Oil production in the United States peaked in 1970, and since then industry analysts have been debating just how long global oil reserves will be able to meet the growing demand. Some analysts believe that global production is reaching its peak right now. You can read about the facts and figures regarding global oil reserves in many articles such as this, but the truth is that no one really knows how much oil is left that can be feasibly extracted. OPEC producers keep it a big secret, as they don't want to disclose how much oil they have found or think they will find. Some people think that the

world is rapidly reaching the point where the growth in new supplies of oil cannot keep up with the pace of oil depletion. Officials in the oil industry agree that the days of major new finds of cheap oil in well explored countries such as the US are over. There is still plenty of oil, but it is getting ever harder to find. However, as oil extraction technology develops, many prospects that were not economically feasible to develop before are now being developed. In the future, there will be prospects that are not worth developing today that future technologies will be able to handle. For instance, horizontal drilling is a new technology that means new pockets of oil that were previously unaffordable can now be extracted. In addition to making extraction more feasible, advancements in technology is helping in the search for oil. New fields are



being found in areas that were once thought to be fully exploited. For instance, 3D visualization technology now enables geologists to view underground formations to a much higher degree of clarity and detail than before. Whilst the limits of fossil fuel exploration keep being expanded all the time, the fact still remains that the supply is finite. Technology is enabling companies to find more oil, but the ever harder methods of extraction are ultimately going to bring the price of fossil fuels up to the point where alternative methods energy production become much more viable. This can only be a good thing. Far from the energy crises, and the 'return to the stone age' scenario that some doomsayers predict, the ending of the world's reliance on fossil fuels which be a huge step for our civilization. Although this is not a strict rule, generally the world's most successful economies have got to where they are because of the major fossil fuels of oil, gas and coal. In the future, the world's most successful economies are likely to be the ones that have embraced the 'green revolution,' and who have developed significant sources of green energy. As has been discussed, with the development of new technologies for exploration and extraction, no one knows quite how long fossil fuels will be economically feasible, but without doubt every country in the world right now should be preparing for this eventuality.

The Stone Age scenario that some doomsayers predict, the ending of the huge step for our civilization.

Although this is not a strict rule, generally the world's most successful economies have got to where they are because of the major fossil fuels of oil, gas and coal. In the future, the world's most successful economies are likely to be developed.

- MAGAZINE TEAM

Manage Your Work, **Don't Let It Manage** **You: Tips for Managing** **Your Time**

Goals of Time Management:

1. To be able to have control over your life - manage your time, don't let it manage you.
2. To be healthier and happier (less stress).

By now, second semester is well underway for most of you, which means a new set of classes with new professors, readings, papers, and exams. After a much needed winter break, it can seem overwhelming to be thrust back into the college balancing act of work, social life, and academics, but some of you learned first semester (some not so much) that time management is key to success. Those of you who did well first semester will need to rewire your brains into academic mode, and those who didn't do so well will need guidance on how to better manage your time so you're not scrambling at the last minute. Here are some tips for studying and time

management that you (or anyone!) can use as a reminder or to help change stress inducing habits.

Things to DO:

1. *Don't procrastinate (the big one!):*

Give yourself enough time to get everything done. Work always suffers when you try to cram in everything at once, as numerous studies prove there's a strong correlation between time management and academic performance. Despite the evidence, some insist they work better under pressure, but most of the time those students have never tried tackling anything ahead of schedule. Just imagine what could be accomplished if you take the time to map out a schedule and stay on top of important tasks. You'd probably feel a lot less stressed and complete your work more thoughtfully.

2. *Break things up into manageable pieces and prioritize:*

The prospect of writing three term papers can be overwhelming, so it's important to come up with a workable plan. Make a to-do list, including deadlines. Then tackle the most important and urgent assignments first.

3. *Face your anxiety and conquer it:*

Remember that a small dose of anxiety can make you buckle down and put on your game face. Cramming for an exam is bad, but not studying at all is worse. Stressing just enough to feel more motivated, determined, and focused can be a real plus. But it's important to take action when stress starts to disrupt other parts of your life

or hinder studying, which brings me to my next two tips.

4. *Talk to your Professors:*

Don't be afraid to raise the white flag. Sometimes, making a plan for meeting academic demands helps demonstrate that it's simply not possible to meet all of the deadlines. Allow yourself enough time to consult with your professors if you're finding it tough. Most professors prefer that students ask for support before blanking on a test or freezing during a presentation. If you seek help after the fact, it may be unclear to the professor whether you're being sincere or just trying to make up for not being prepared. And the relationships you establish by seeking help are just as important as extra study sessions.

5. *Talk to a counselor:*

Counselors at your school's counseling and academic skills centers can lend a hand in managing the triggers that cause unnecessary stress. Scheduling a one-on-one appointment, joining a discussion or study group, or simply learning a few new strategies through an informal conversation can be helpful. And the relationships you establish by seeking help are just as important as extra study sessions.

6. *Bag the booze:*

There's a reason they call it getting wasted. Alcohol dulls motivation, lessens productivity, and causes students to waste time. If you already don't want to write that paper, skipping it won't seem like a big deal after a few drinks. In the morning, though,



you'll be cursing yourself for not doing it.

7. Take a break from distractions and stay focused:

While your computer is full of all sorts of fun diversions – instant messaging, social networking sites, movies, and more – turn it off when you're trying to focus on school work. You may think you can handle writing a history paper and chatting with a friend, but your mind will be better off if you focus on the task at hand. Save the socializing for study breaks.

8. Follow your instincts:

If there's a voice inside you saying, "I really shouldn't be going out tonight," you should listen to it. If you're going to feel guilty for going out, you won't have that much fun anyway.

9. Study for your hardest classes first and study often:

If you can't remember someone's phone number at the bar without writing it down, you're not going to master European history overnight. This also goes back to prioritization – if you leave the hardest classes for last, then chances are you might not even get to them.

10. Designate study days and party days:

No, "I only drink on days that end in y" doesn't count. If you decide that your weekend starts at Thursday's happy hour, commit the rest of your week to hardcore studying. Just

remember that the sooner you get your studying commitments out of the way, the sooner you can have fun.

11. Catch some Z's:

At some point, we all need a break. Sleep will help you remember the things you read and help you process ideas and arguments for a paper. Having a regular sleep schedule will make a world of difference in how you do academically.

12. Check out your school's exam policy:

A lot of schools limit students to two exams per day. If you're over the limit, you may be able to postpone some.

- MAGAZINE TEAM

London offers lessons for future

Olympic organizers got several things right, but made a few mistakes. Here's what was good and where they erred.



London- with just hours left to go in the 2012 London Olympic games; one can safely say that this has been an utterly enjoyable edition. World records began toppling from the first day of events, the opening ceremony

was more-or-less widely appreciated, the home team has done spectacularly, the city of London has coped courageously with the record crowds, and almost every high-profile discipline has delivered memorable moments. But what were some of London's finest achievements? Where did they mess up? What should future host cities learn from London?

Do's:

1. There are two secrets to how London coped with the influx of visitors. The first is exceptionally thoughtful utilization of the city's public transportation system. Thanks to major sporting events such as the premier league of football, the city is no stranger to handling sudden bursts of traffic as stadiums fill up and empty out. London combined those learning's with robust online routing websites, extensive signage and immediate repairs of breakdowns on the tube network. But the masterstroke was giving away an all access day pass to the city's transport system with every ticket to an Olympic event.

2. The second secret was astonishingly friendly network of volunteers. It was well nigh impossible to go anywhere in central London or around the venues without bumping into an Olympics volunteer. Look even mildly disoriented and a volunteer would pounce on you immediately. What is even more remarkable: almost all of them were unpaid.

3. London didn't go overboard on security. Before the games, there were fears that grim armed men in uniform

would descend on the city. Who knows? Perhaps they did. But it wasn't that obvious. There were also fears that security checks at venues would lead to long lines. Instead, the organizers executed an airport-like strategy at venues such as the Excel Arena. Early arrivals could spend time at the restaurants, watching exhibitions or watching events on the big screen. Leaving the security areas free for latecomers.

Don'ts:

1. This is 2012. The world deserves a better online experience to view the Olympics especially in countries saddled with rubbish broadcasters such as NBC. The IOC needs to think about scrapping nation-by-nation rights to streaming video and having one globally accessible video portal.

2. Don't scare away people. Especially the locals. Especially if your transport systems have been sorted out. There have been several reports, some disputed, that contrary to expectations, the Olympics have led to lower traffic and retail business in central London. In other areas, crowd management has been so efficient that traders who paid nobody to sell to. Host cities need to turn down the paranoia a little.

3. Empty seats are inexcusable. Especially when millions of applications from British buyers, whose taxes pay for the jamboree, have been turned down. The organizers did try to sell some unused tickets back to the public. But it was too little too late. This writer spotted two entire vacant sections during the women's football



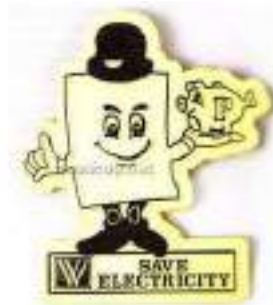
gold medal match. The IOC needs to rethink its policy of reserving seats for the so-called "Games Family".

4. Boxing needs to clean up or ship out. Badminton had its problems during the games. But at least they had the gumption to do something about it. Boxing continues to bewilder.

- MAGAZINE TEAM

Emerging green technology:

***"TODAY'S WASTAGE IS
TOMORROW'S SHORTAGE"***



Natural News Researchers from the University of Minnesota's (UM) College of Science and Engineering are working with a new alloy material that they say is capable of turning waste heat -- like the kind emitted from vehicle exhaust pipes or from air conditioning units -- directly into electricity. Though still in its infancy, the technology has the potential to revolutionize the way heat is recycled, and it may one day offer individuals the ability to recycle an unlimited amount of heat into free energy.



This research is very promising because it presents an entirely new method for energy conversion that's never been done before," said Richard James, head of the research team and professor of aerospace engineering and mechanics at the college. "It's also the ultimate 'green' way to create electricity because it uses waste heat to create electricity with no carbon dioxide." Published in the new scientific journal *Advanced Energy Materials*, the findings explain how the scientists developed their new multiferroic alloy, known as $\text{Ni}_{45}\text{Co}_5\text{Mn}_{40}\text{Sn}_{10}$, which involves heat absorption caused by rapid transformations between solid states, which in turn produce electricity. Or to put it more simply, the unique alloy material absorbs heat when it morphs, and in generates electricity.

Multiferroic processes are not necessarily new, but the kind developed by the UM scientists minimizes a process called hysteresis, which basically infers the loss of heat energy during its conversion to electricity. By minimizing hysteresis, the scientists were able to capture and conserve breakthrough amounts of energy from their unique process."This research crosses all boundaries of science and

engineering," added James. "It includes engineering, physics, materials, chemistry, mathematics and more. It has required all of us within the university's College of Science and Engineering to work together to think in new ways."

- MAGAZINE TEAM

The Countdown of the APOCALYPSE (Dec 21st 2012) is on

The Mayan calendar of 2012 comprises of range of eschatological beliefs according to which cataclysmic or transformative events will occur this December or to be more specific on the 21st December. This very data is allegedly marked as the end of an era that would be reset the data to zero and signal the end of the humanity. This data is regarded as the end date of a 5125 year long cycle in Mesoamerican long count calendar.

A new age interpretation of this transition is that the date marks the start of time in which earth and its inhabitations may undergo a positive physical transformation that the beginning of a new era.

Other suggest that the 2012 date marks the end of the world or a similar catastrophe, with theories suggesting the end of the world include the arrival of the next solar maximum on earth's collision with a black hole.

As per the Mayan Calender, which many scholars believe to the most accurate calendar in the

world, it is predicted that a final event that includes a solar shift ,a Venus transit and violent earthquakes to occur.

According to this calendar the world will end after 13 b'ak'turnsthat is roughly 5125 years. Although this through of a catastrophe which will destroy the world is believed by many common people present scholars suggest otherwise.

As a famous scholar, SVEN gronomeyer says "human beings seem to be attracted by apocalyptic ideas & always assume the worst". We human beings are so much into the ideas of thinking and predicting the future that any theory fed to them is believed & is also portrayed as something very integral in our lives. So as of now, the biggest question that arises is whether to believe the Mayan calendar theory or not. Recalling back the instaus in which many such theories have failed to happen.so,there is a propobility that this theory could also be a HOAX.So,as per the prediction for the world to end or not, all we can do is wait till the Dec'21st,12 and hope for the best to happen.

**-R.DEEPIKA
ECE-B
Final year**





CHAT BOX :

**Interview with Mr. E. Alwar,
Assistant General Manager of
Personnel (H.R.) of RAMCO Cement
Factory.**

Q1: What are the basic requirements of the candidate who apply for the job in your company?

A: The candidate who is applying for the job in our company should score minimum marks as 60% in every semester (at 1st attempt).

Q2: There are many types of interviews, for example, written test, group discussion, face to face interviews. Likewise how many types of interviews happen in your company?

A: In our company 3 types of interviews are taken, they are written test, personal interview and group discussion. Written test consists of general aptitude which is a one hour test. Then the qualified students will be called for personal interview and it will be taken by the HR of the company. The candidates those who qualify in the personal interview are then called for the final test.

Q3: How much time do you take to interview a candidate?

A: The time duration will be according to the candidate's ability to answer. But, normally it will be for 15-30 minutes.

Q4: What is your opinion about the importance of ECE engineers in cement industries?

A: The ECE engineers have more importance in every company but compared to other industries, there are very less scope for the ECE students in cement industries. In cement industries ECE students have high demand only in the manufacturing sector.

Q5: How much percentage of ECE engineers work for your company?

A: Only 5% of ECE engineers work for our company because, very well equipped machines are used in the manufacturing sector which is fault free and so there is a little requirement of ECE engineers.

Q6: What is your opinion about the importance of ECE engineers in other companies?

A: Normally, in other companies there are more number of ECE engineers compared to other discipline of Engineering.

Q7: How many candidates are called for an interview for a single vacancy?

A: We call all the applicants who fulfill the following criteria:

- a) The educational Qualifications should be related to the Job vacancy.
- b) The candidates should score minimum of 60% at first attempt.

Q8: Whether the extracurricular activities and the International

certifications are considered in your industry?

A: We do not give much importance to extracurricular activities but we consider International Certification courses. But, IT industries give more importance to these things.

Q9: Do you prefer students with 80% marks having no interest in sports or those with 65% marks and good consistent record in sports?

A: In our company we will not give much preference to sports but, we will consider good overall percentage and depth of knowledge. So, if we have to choose between 65% and 80%, we will give preference to 80%.

Q10: What do the selected candidates do in the training session of your industry?

A: The selected candidates get to know about the company and the working of the GM, AGM, MD, etc. and also they have hands-on-session in the manufacturing sector.

-The interview was taken by

*N.T.Theejhasri
II yr (ECE-D)*



TICKLE YOUR BRAIN

From the clues given find out the names of **CARS** :

- 1) A Japanese Fighter
- 2) A Zodiac Sign
- 3) Speedy; Quick
- 4) A Thickly Populated Urban Area
- 5) A Soldier with a Lance
- 6) A Country's Representative to another Country
- 7) To have a High Regard
- 8) A Sight Seeing Trip to see animals in their Natural Habitat
- 9) People who live in Caravans
- 10) A Color present in VIBGYOR

For answers see Pg. no-89

SKETCHES



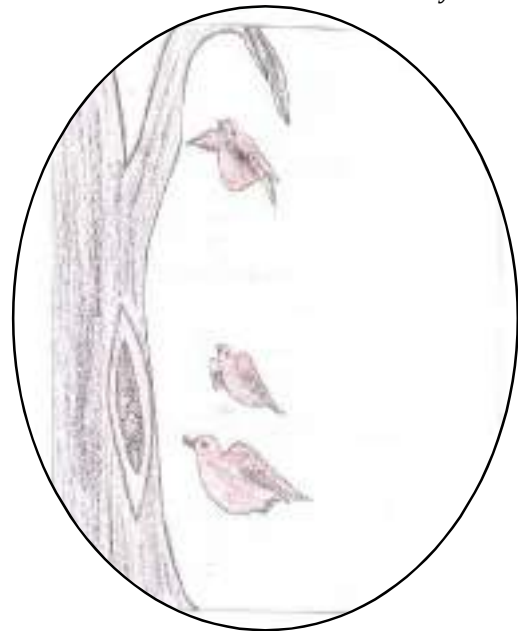
R.EUREKA SRI
ECE-A
Prefinal year



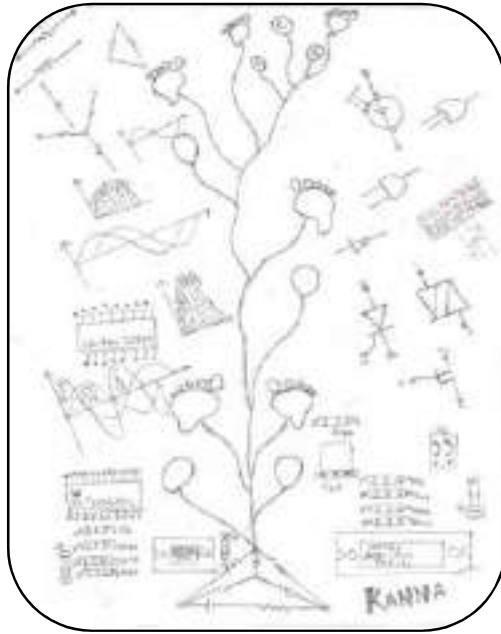
-DEVI JAYA THANGAM.T
ECE-B
Final year



DHAMOTHARAN.M
ECE-A
Prefinal year



HARINI.B
ECE-B
Final year



-VIGNESH KANNA
ECE-D
Second year



-EMILIUSSCINDIA.D
ECE-B
Final year



-M.DAISY
ECE-B
Final year



-M.LEKSHMI
ECE-E
Second year



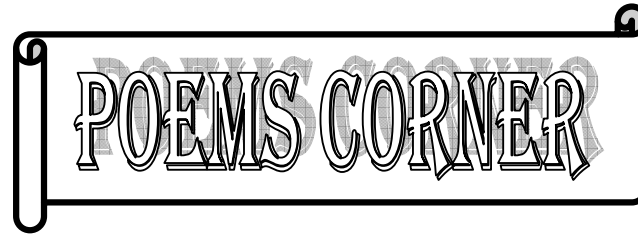
-VIGNESH KANNA
ECE-D
Second year

BEFORE MARRIAGE



AFTER MARRIAGE

-M.LEKSHMI
ECE-E
Second year



Don't Quit

Anonymous

*When things go wrong as they
sometimes will;
When the road you're trudging seems
all uphill;
When the funds are low, and the debts
are high;*

*And you want to smile, but you have
to sigh;
When care is pressing you down a bit
Rest if you must, but don't you quit.*

*Success is failure turned inside out;
The silver tint of the clouds of doubt;
And you can never tell how close you
are;*

*It may be near when it seems afar.
So, stick to the fight when you're
hardest hit -
It's when things go wrong that you
mustn't quit.*

Contributed by Prof. VK. MADAN

*This hurts our planet in every single
way.*

*The changes could leave us all in
sorrow and dismay.*

*We need to stop it now so that the
temperature doesn't rise.*

*People, plants and animals would be
otherwise in demise.*

*Changes in temperature due to
depleting ozone layer.*

*We really don't need it so let's show
that we care.*

Mother Nature can't do it all.

So let us give her at rest.

We all need to try and do what is best.

*Our planet Earth is precious and can't
be replaced.*

*We need to act now or our home will
be erased.*

*-KHUSHBOO
ECE-C
Final year*

Global warming

Global Warming isn't hard to explain.

*It leaves Mother Earth crying with
excruciating pain.*

Mother is a God Gift:

*First word a child learns to speak is
mother,
Mother is the best part of our life,
Mother takes so much pain,
Mother keeps her child for 9 months
in one part of her body,
But then to she never complained to
her child,*



*Mother bears every pain to protect
her child,
If there would be small amount of
food then she will not
But make her child eat that food,
Mother will always fight for her
children's happiness,
Never hurt your mother because
She will never tell her child that she is
hurt,
Mothers are the second face of God..*

M.VIGNESH
ECE-D
Second year

NATURE:

*Bringer of life, majestic deity
Worthy of the praise Of all humanity
You inspire and daze, And make us
feel free.
Always vigilant, Over your domain
Nursing every plant, Back to life
again.
Creating new forms, And brightening
our days
After rain and storms, With your
magic rays.
Warmer of the earth, And of orbs
beyond
Forger of rebirth, In the cosmic pond.
Keeper of creation, Before the world
begun.
Deserver of adoration, our
everlasting Sun.*

M.VIGNESH
Second year ECE-D

Answers:

1.Tata Sumo 2.Mahindra Scorpio
3. Maruthi swift 4.Honda City 5.Lancer
6.Ambassador 7.Maruthi Esteem 8.Tata
Safari 9. Maruthi Gypsi 10. Tata Indigo

--**M.LEKSHMI**
ECE-E Second year

Times Are Changing

*We had our good days and we had
our bad,
I thought I would never forget the
times we had
The times are changing
and our memories are slowly
fading
and though times may change
we just have to hope and pray
that someday thing will be okay
the times are changing
but we can't keep holding onto the
past,
because we don't know if our past
can be part of our future
and as I shed one last tear
I live in fear
that as the times change
I will lose you forever.*

M.VIGNESH
ECE-D
Second year

MOTHER POEM

*There are times only when a
Mother's love
Can understand our tears,
Can soothe our disappointments
And calm all our fears.*

*There are times when only a
Mother's Love
Can share the joy we feel
When something we've dreamed
about
Quite suddenly is real.*

*There are times when only a
Mother's faith
Can help on life's way*

*And inspire in us the confidence
We need from day to day.*

*For a Mother's heart and a Mother's
faith*

*And a Mother's steadfast love
Were fashioned by the Angels
And sent from God above...*

My Day!

*I always try my best,
To prove that I am better than the
rest.*

*But don't know what goes wrong,
Making my life a doleful song.*

*I am all alone in this world,
And the mistake is what I was not
told.*

*I don't want the way it's going on,
I cannot wait till dawn.
I miss the good days that I ever
had,*

*And miss the affection of my mom
and dad.*

*Lastly, there's just one choice to
have patience,*

*Until my good luck comes back
once.*

*But there's light still shining out,
I was helped by my friend, no
doubt!*

*That is all I can, say,
Eagerly waiting for 'My Day'.*

**-KHUSHBOO
ECE-C
Final year**

A woman is...

*Mesmerized by her liquid eyes
The ancients called her a deity,*

*Scared of her immense power
Some stood before her to worship.*

Unmatched by her love and care

She is adored as a mother,

*A playmate as a sister and a soul
mate as wife,*

*She wipes everybody's tears and
shares everybody's grief*

*To many she is a silhouette of an
hour-glass,*

*But when the hour is divided into
minutes, seconds*

*And minute moments, she occupies
the eternity,*

*Transcending base desires, lust and
human frailty
A woman is not just a
showcase of a home,*

*A bath tub doll, but a life-line of the
creation,*

*She is not the fuel to keep the passion
burning
But the morning rays to keep the days
glowing!*

A woman is a flower of the paradise,

*Ever to remain fresh and never to
wither,*

*Ever to spread the fragrance of love
And be the source of infinite pleasure*

**-BOYINA SANTHOSH
ECE-B
Final year**



Silence to words And Words to silence

*At present,
I am struck in situation
That forces me to hear
That forces me to speak
That forces me to obey in fact
But even when forced
She, the little one did not
While I speak with pleasure
She is not even able to whisper
While I hear to get excited
She is belonging to hear
While I mean words
She means expressions
And all these made me
Only to be spell bound
And watch with silence
The actions of cute little sister
Which made me feel that?
Silence is more valued.....*

-P.FATHIMA SANOFER
ECE-B
Final year

My cute brother

*A little younger than me
But a little wiser than me*

*Always a step behind me
Not because you are slow
Just because you are young
Always a little faster than me
Not to put up a show
Just to take me forward
And all those you do
Takes the word selfish
And means the word- brother*

-S.KARTHIKEYAN
ECE-B
Final year

Teacher

*Greet your teacher
Who builds your future
Be a friend of teacher
They will fill you with food feature
They are like your parents
Who will give you a number of talents
Teachers are those priests
Who show you the way to Christ
Their scolding are that love
Which makes your life easy from the tough?
They are those judges
Who never teach you to be a budge?
Teachers always teach you truth
Which makes your life full of fruit?
So teachers are the only one*

*Who spend their lives to make your
life full of joy*

-H.GAAYATHRI
ECE-B
Final year

YET TO WIN

My cause is just the one

Who thinks only one

But have never won

Would have been better with an iron

Instead of flash that with an iron

May be the power of dictation

With which if I could have born

Would have made people see

Better than what I have shown

At least I an gone for all.....

-S.KARTHIKEYAN
ECE-B
Final year

Be a friend

Be a friend .you don't need money

Just disposition sunny

Just the wish to help another

Get along some way or other

Just a kindly hand extended

Out to one who's unbefriended

Just the will to give or lend

This will make you someone's friend

.Be a friend. The pay is bigger

(Though not written by a figure)

Than is earned by people clever

In who's merely self-endeavor?

*You will have friends instead of
neighbors*

For the profits of your labors

You will be richer in the end

Than a prince, if you are a friend

-SAMPADA.Y
ECE-E
Second year

Secret of success

*"What is the secret of success?" asked
the sphinx*

"Take pains", said the window.

"Always keep cool", said the ice.

*"beep to date", said the calendar
"Never lose your lead. "Said the
barrel.*

*Make light of everything", said the
fire.*

*"be sharp in all your dealings. "said
the knife.*

-SARAVANAN.C.S
ECE -E
Second year

Books

*Simply great
With exceptional wealth*

*Always at the top
In helping people top*

*Purity, its power
Clarity, its cause*

*Perfectly predictable,
That surely worth able*

*And a perfect gain
As it relieves pain*



-S.KARTHIKEYAN
ECE-B
Final year

Days passed

Sunday-the day of wait

Monday-just a bit satisfied

Tuesday-seems a bit distant

Wednesday-busy but still bothering

Thursday-enriched though troubled

Friday-something to cheer

Saturday-a better day at last

But tomorrow is Sunday again

-M.DURAIRAJ
ECE-B
Final year

Achieve

If you want to enjoy just

Think today is the first day

But,

If you want to achieve

Think today is the last day

-NAGARAJ.S
ECE-E
Second year

QUOTES :

1. Brilliant acts, Idiot reacts.
2. Death can make an end to your life not to your achievements.
3. Think twice to do it wise.
4. Mother is a thesaurus of love.
5. Friendship is a tool to shape your life not to save your life.

6. I can't be like you But I can be a past of life to be with you.

7. Experience is word that is derived from failures.

8. Friendship doesn't decide your character it shows your reality. 9. Live with your desire don't male trust on your destiny.

10. Justification makes dissatificaton.

JEROME BOB ROY.M
ECE-B
Final Year

HINDI POEMS

समय का खेल (Time)

समय गुजरता जा रहा है,
मंजिलो को भुलता जा रहा है,
क्या समझाऊ अपने आपको,
दिल भी भटकता जा रहा है।
एक पल सोचू तो दूजा पल याद आता है,
स्वप्न,स्वप्न ही रह जाता है,
किस्तों भरी इस दुनिया में यारा,
अपने अपने को भुलता जा रहा है।
समय को देख क्यों रो रहे हो,
पतझर में भी बीज क्यों बो रहे हो,
इस नस्वर भरी दुनिया में भी,
जागकर भी क्यों सो रहे हो।
काल की बातें काल समझ रहा है,
प्रातः काल की बातें अन्धकार बुझ रहा है
इस भागमभाग की जिंदगी में भी,

जागने वालो को भी सिर्फ कमाई सूझ
रहा है

समय गुजरता जा रहा है,
मंजिलो को भुलता जा रहा है॥

-MUKESH KUMAR ARYAN
ECE -F
Final year

वक्त नहीं! (No time)

हर खुशी है लोगों के दामन में
पर हँसने का भी वक्त नहीं,
दिन रात दौड़ती दुनिया में
जिन्दगी के लिए ही वक्त नहीं।
माँ की लोरी का ओ एहसास है
पर माँ को माँ कहने का वक्त नहीं,
सारे रिश्तों को हम मार चुके हैं,
लेकिन उन्हें दाँ नाने का भी वक्त नहीं
सारे नाम मोबाइल में हैं
पर दोस्तों के लिए वक्त नहीं,
गैरों की क्या बात करें,
जब अपनों के लिए ही वक्त नहीं।
आँखों में है ख्वाब कई
पर उन्हें पूरा करने का भी वक्त नहीं।
पैसों की दौड़ में ऐसा दौड़े,
की रुकने का भी वक्त नहीं,
पराए एहसासों की क्या क्रूर करें,
जब अपनों के लिए ही वक्त नहीं।
तू ही बता मेरी जिन्दगी
इस जिंदगी का क्या होगा
कि हर पल मरने वालों को
जीने के लिए भी वक्त नहीं ॥

-Khushboo
ECE-C
Final year

माँ की याद (Memory of Mother)

माँ के हाथों की बनी जब दाल रोटी याद
आई,
पंचतारा होटलों की शान शौकत कुछ न
भाई,
बैरा निगोडा पूछ जाता किया जो मैंने कहा
सलाम झुक-झुक करके
मन में टिप का लालच रहा,
खाक छानी होटलों की, चाहिए जो ना
मिला
क्रोध में हो स्नेह किसका?
कल्पना से दिल हिला
प्रेम मे नहला गई जब,
जम के तेरी ांट खाई,
माँ के हाथों की बनी जब दाल रोटी याद
आई,
तेरी छाया मे पला सपने बहुत देखा किए ,
समृद्धि सुख की दौड़ मे दुख भरे दिन जी
लिए
महल रेती के संजोए शांति मैं खोता रहा ,
नींद मेरी छिन गई , बस रात भर रोता
रहा ,
चैन पाया याद करके लोरी जो तूने सुनाई
माँ के हाथों की बनी जब दाल रोटी याद
आई /
लाभ हानि का गणित ले जिंदगी की राह
में ,
जुट गया मित्रों से मिल प्रतियोगिता की
दाह में
भटका बहुत चकाचौंध में खोखला जीवन
जिया ,
अर्थ ही जीने का अर्थ,
अनर्थ में पुबो दिया
हर भूल पर ममता भरी तेरी हँसी सुकून
लाई,



माँ के हाथों की बनी जब दाल रोटी याद
आई /

-SUDHANSHU KUMAR
ECE-E
Final year

उम्मीद के बंद लिपों में

(Hope)

उम्मीद के बंद लिपों में एक
टिमटिमाता सपना है,
बादल की पिचकारी लेकर चाँद सितारे
रंगना है ,
सुरज की लालिमा सा,

एक उज्ज्वल भविष्य बनाना है /
दुनिया की इस भीड़-भाड़ में ,
अपनी एक पहचान बनाना है /
खुशियों के हरेक लम्हे को दोस्तों संग
बिताना है
दुखों की काली चादर को हमेशा क लिए
मिटाना है जीवन के हरेक अनुभवों से
हमेशा कुछ सीखना है उम्मीद के बंद
लिपों में एक टिमटिमाता सपना है /
बड़ी बड़ी हस्तियों जैसी ऊँची इमारतें
बनाना है परिवार के उमीदों की
दिव्यज्योति पे खड़ा उतरना है /
भगवान की इस जीवनदायी को
एवरेस्ट तक पहुँचाना है ,
प्रकृति की हरियाली सा
खुद को हराभरा बनाना है
तूफान की रफ्तार सा खुद को तेज बनाना
है,
धीरुभाई अम्बानी जैसा ,
सपना लता की सीढियाँ चढ़ना है /
अपना अभिमान को कभी ना कम होने

देना है /
आसमान की किताब पर
बस नाम अपना लिखना है,
उम्मीद के बंद लिपों में एक
टिमटिमाता सपना है ,
और इस सपने को साकार करना है //

-SWETA KUMARI
ECE-F
Final year

Hindi Quotes

सपना लता का राज (Secrets of Success)

१. कार्य उद्यम से ही सिद्ध होता है, मनोरथ से नहीं ।
२. प्रत्येक अच्छा कार्य पहले असंभव नज़र आता है
३. उद्यम ही सपना लता की कुंजी है।
४. एकाग्रता से ही विजय मिलती है।
५. कीर्ति विरोचित कार्यों की सुगंध है ।
५. भाग्य साहसी का ही साथ देता है।
६. सपना लता अत्यधिक परिश्रम चाहती है।
७. विवेक बहादुरी का उत्तम अंश है।
८. संकल्प ही मनुष्य का बल है।
९. कर्म करने से ही अधिकार है, ल से नहीं।
१०. अपनी शक्तियों पर भरोसा करने वाला कभी असपना ल नहीं होता है।

-ANISH ANAND
Second year ECE-A

TAMIL POEMS**மயக்கம்**

இந்த நாட்களிலே.....!

நினைத்துப்பார் நிதானமாய்
நிம்மதி பெருமூச்சுவிட்டு...!
கழிந்துபோன காலங்களை
கானலாய் போன உறவுகளை...!

எந்திர மயக்கம்.....!

செல்லிடப்பேசியின் மயக்கத்தால்
செலவழிக்கப்பட்ட தொகையினில்...!
அடிமையாகிப்போன பழக்கத்தால்
ஆளுமை இழந்த செயல்கள்...!
விரயமாக்கப்பட்ட நிமிடங்களால்
விணைக்கப்பட்ட சிந்தனைகள்...!
எத்தனை...! எத்தனை...!
நினைத்துபார்.....

வாலிப மயக்கம்.....!

ஊடல் என்னும் மயக்கத்தால்
ஊறிப்போன நெஞ்சங்களில்...!
உறிஞ்சப்பட்ட உணர்வுகளால்
உதறப்பட்ட சொந்தங்கள்...!
சுரக்கப்பட்ட கண்ணீர்களால்
சுனையற்றுபோன நினைவுகள்...!
எத்தனை...! எத்தனை...!
நினைத்துபார்.....

இறுதியாக.....!

மயக்கம் மதியிழக்கச் செய்யும்
தயக்கம் தவறிழைக்கச் செய்யும்
தயக்கமின்றி மயக்கம் விடுத்தால்
தலைநிமிர்வாய்! நீ தலைநிமிர்வாய்.....!

-மா.பரமாத்மா
உதவி பேராசிரியர்

என் வாழ்வில்**தொழில்நுட்ப வளர்ச்சி**

அன்னை அழைத்தும் எழ மறுத்த
நான்!!!!
அலாரத்திடம் அடங்கி போகிறேன்....
நீர் அலையில் ஆடிய நான்.....
நித்தம் அலைபேசி வலையில் சிக்கித்
தவிக்கிறேன்!!!!
மண்வெட்டி பிடித்து மடை திறந்த
காலம் போய்....
இன்று மடிகணினி வழியே தான்
மனம் திறக்கிறேன்!!!!
தறிநெய்து உடை உடுத்திய
நான்.....இன்றோ
தறுதலை உடை வழக்கத்தை
விரும்புகிறேன்!!!!
இயந்திரம் தரும் சுடுநீரையை
தேநீர்,காபி என்று நம்புகிறேன்....
தாய் தரும் தண்ணீரையும் உற்று
நோக்கி பின் அருந்துகிறேன்!!!!
சொந்த பந்தங்களை தொல்லை என்று
கூறி வந்த நான் இன்று....
முதல் முதலால் சந்திக்கும் நபரையும்
அங்கிள் என்று அழைக்கிறேன்!!!!
பட்டம் விட்டு காற்றின் திசை அறிந்த
நான்....
கடற்கரை திசை கேட்டு காற்று
வாங்க பயணிக்கிறேன்!!!!
இத்தனைக்கும் மேலாக காகிதத்தில்
என் கவிதையை கிறுக்காமல்.....
காலத்தை கருத்தில்
கொண்டு,மின்னல் வழியே
தங்களுக்கு எழுதி மொழி மாற்றம்
செய்கிறேன்!!!!

வி.முருகானந்தம்
இரண்டாம் ஆண்டு 'இ' பிரிவு



ஜெய்ஹிந்த்!!

தாயின் பாசமும், தாய்நாட்டின்
சுவாசமும்;
தாயின் மனமும், தாய்நாட்டின்
குணமும்;
தாயின் அன்பும், தாய்நாட்டின்
பண்பும்;
ஒருங்கினைந்து முழு மனிதனாய்
இவ்வுலகில்,
உன்னை நிலை நாட்டியது.....
தாயிற்கு சேவை செய்தோம்,
தாய்நாட்டிற்கு என்ன செய்தோம்????
சிந்தனை செய்!!!
இச்சுதந்திர நன்னாளில் ஏதெனும்...
சிறிய நற்செயலை செய்ய "துணிந்து
வா...!!
சிறந்த குடிமகனாக மட்டுமல்லாமல்
சிறந்த சேவகனுமாய் வா...
தாயை பெருமிதம் கொள்ளச்
செய்வோம்!!
தாய்நாட்டை முன்னேற்றம் அடையச்
செய்வோம்.....!!

சோ.சந்திரமோகன்
இறுதி ஆண்டு, 'பி' பிரிவு

கல்லூரி

பயம் தந்த பெரிய மதில் சுவர்கள்
நடுக்கம் தந்த உரிய கட்டிடங்கள்
கல்வி தந்த தனித்திறமை
ஆசிரியர்கள்
அறிவு தந்த நூலகப் புத்தகங்கள்
பரிசு தந்த மேடைகள்
வெற்றி தந்த விளையாட்டு விழாக்கள்
நடுக்கம் தந்த ஆங்கிலப் பாடங்கள்
உரிமை தந்த நாய்காலிகள்
உறக்கம் தந்த காற்றடிகள்
அடுத்தடுத்து வந்த வாரத்தேர்வுகள்

வடை தந்த கேண்டின்
நிழல் தந்த மரங்கள்
செய்தி தந்த தகவல்பலகை
நற்பண்புகள் தந்த பாடப்புத்தங்கள்
மகிழ்ச்சி தந்த மணியோசைகள்
அன்பு காட்டிய நண்பர்கள்
இவையனைத்தும் கூறின!
இன்னும் சில நாட்களே உன்னுடன்
என்று....

ம.ஹரிகிருஷ்ண மூர்த்தி
இரண்டாம் ஆண்டு

ஆறாம் அறிவு

தொழில்நுட்பம்

கிழிந்த ஓரமும்
அழுக்கு சட்டையும்;
அடுத்த வேளை
உணவுண்டா அறியா!!
காய்ந்துப்போனச்
சிகையும் வயிறும்;
நின்று களைத்துப்போன
முதுகெழும்பும்
சிணுங்கிக்கொண்டு;
படுக்கத் தேடும் இடம்;
சாலை ஓரமா;
இல்லை ஓரத்தில்
உள்ள மரமா நானறியா!!
சோகத்திற்குச்
சுண்ணாம்பு அடித்து;
பறிக்கும் புன்னகைச்
செய்யும் நான்தான்
செல்வந்தன் இன்று!!!!

ச.காளிராஜன்
மூன்றாம் ஆண்டு, 'சி' பிரிவு

தோழன்

உன் முகத்தை பார்க்க

உன் தாய்மகள்
நீ கருவறையில் இருந்து
வெளிவரும்வரை காத்திருப்பாள்!!!
ஆனால்,
உன் முகத்தை பார்க்க
உன் கல்லறையிலும் காத்திருப்பான்
இந்த தோழன்!!!!

சி.வீரனன்
இரண்டாம் ஆண்டு

நம்பிக்கை

செடியைவிட்டு மலர் பிரிந்தாலும்
சிரிக்கிறது ஏனென்றால்.....
பிரித்தாலும் மீண்டும் மலர்வேன்
என்ற
நம்பிக்கையில்....
உன்னை விட்டு வெற்றி பிரிந்தாலும்
நம்பிக்கையுடன் இரு....
பெரிய வெற்றி நீ அடைய போகிறாய்
என்று..!!

வி.முருகானந்தம்
இரண்டாம் ஆண்டு, 'இ' பிரிவு

கவிதை துளிகள்

முக மூடிகள் மிக அழகானவை....
அவரவர் சுயரூபத்தைவிட!!!!
கையில் அரிவாள் முறுக்கு மீசை
மிரட்டும் விழிகள்.....
முனியப்ப சாமியிடம் நலம்
விசாரிக்கும்
ஊர்க்குருவி தோளில் அமர்ந்தபடி!!!!
எதிர்பாராமல் பெய்து
இதம் தந்த கோடை மழை
ஆனாலும் சபிக்கிறேன்
மைதானத்தில்
விளையாடும், குழந்தைக்காக!!!!

முன்வழியே பயனித்தும்
கிழிபடாமல் செல்கிறது, காற்று!!!!
தலையைத் தானே
வெட்டிக் கொண்டதோ, அரிவாளுடன்
தலையில்லா முனியப்பர் சிலை!!!!
நிழலளிக்க இசைகின்றன
மரங்களின் இலைகள்
பறவை கூடுகளுக்கு.....

சந்திரசேகரன்
இறுதி ஆண்டு, 'பி'
பிரிவு

கல்வி

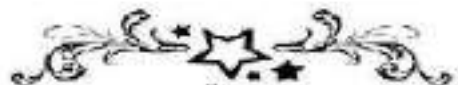
நம்மை ஈன்றெடுத்த தாயை
நேசிக்கின்றோம்
நம்மை வளர்க்கின்ற தந்தையை
நேசிக்கின்றோம்
இவையெல்லாம் நேசிக்கும் நாம்
நம்மை கடைசி வரை வாழவைக்கும்
கல்வியை ஏன் நேசிக்கும்
மறக்கின்றோம்?
தாய் தந்தையைபோல்
நமக்கு நன்மையை மட்டுமே
விளைவிக்கும் கல்வியையும்
நேசிப்போம்!!!!
சாதிப்போம்!!!

மோ.அஷ்வினி
மூன்றாம் ஆண்டு, 'ஏ' பிரிவு

கடன்

கொடுத்துவிட்டு நினைக்காதே....
வாங்கிவிட்டு மறக்காதே..
கொடுப்பதை மறக்காதே..
வாங்கத்தான் நினைக்காதே.....!!!

கார்த்திகேயன்
இறுதி ஆண்டு,
'பி' பிரிவு





TELUGU POEMS

మిత్రమా....మిత్రమా....మిత్ర
మా....మిత్రమా....
(Mitra..Mitra..)

నీధర్యంతోడైఉండగా
ఏసాయంకోసంచూడగ
నీదేయంచూపేమార్గంలోపోరాసూటిగా
ఏనాడూవెనకడుగేయక
ఏఅడుగుతదబడనీయక

నీఘమ్యంచేరేదాకాదుసుకుపోరమిత్రమ
ఏనాడూవెనకడుగేయక
ఏఅడుగుతదబడనీయక
నీఘమ్యంచేరేదాకాదుసుకుపోరమిత్రమ
ఇష్టంఉంటేచేదుకూడాతయనే
కష్టంఅంటేదూదికూడాభారమే
లక్షమంటూలేనిజన్మ ఎందుకురా
లక్షలాదిమందిలేరామందగా
పంతంపట్టిపోరాడందేకోరినవరాలుపొందలేవు
కదా
పట్టుదలతోచేస్తేసమరం
చేస్తూఉంటేవిపనివనసాధ్యమే
చూస్తూఉంటేరోజులన్నీసున్యమే
ఒక్కఅడుగువేసిచుస్తేదాలుర

-BOYINA SANTHOSH
ECE-B
Final year

ప్రేమతొలగేస్తేచింతిస్తుంది.....

ప్రేమనుపం
చినప్రేమనూ
ప్రేమనుపెంచినప్రేమనూ
ఆశగాకోరదాప్రతిప్రసాదయం

ప్రేమనుపొందటంఓవరం
అదిఅంబరమంటినసంబరం
ప్రేమలోతేలుదాంప్రతినిమిషం
ప్రేమలోఈలోకమేసాగెప్రేమగా
ప్రేమతోఈజీవితంప్రేమించగా

ప్రేమనుపెంచినప్రేమనూ
ప్రేమనుపెంచినప్రేమనూ
ఆశగాకోరదాప్రతిప్రసాదయం

పసిమదిలోఏముందోముందుగానేతెలిసుంటుంది
అందుకనేఆధైవంజంటగానేనడిపిస్తుంది
మూసిఉన్నకళ్ళలోఎన్ని ఆశలో
భాషరానిగుండెలోఎన్ని ఊసులో
సిరివెన్నెలంటికున్న హం
గతజన్మలోనిబహుమానం

ప్రేమనుపెంచినప్రేమనూ
ప్రేమనుపెంచినప్రేమనూ
ఆశగాకోరదాప్రతిప్రసాదయం

ఎవ్వరితాఎవ్వరికోప్రేమరాతరాసుంటుంది
ఆమదికీఈమదికీబందమేసినడిపిస్తుంది
గుప్పెడంతగుండెలోప్రేమఅన్నదీ
గ్నాపకాలఊపిరైతాకుంతుంటాదీ

ప్రేమించిచూడుటకసారి
అదిమార్పుతుందినీదారీ
ఈప్రేమలోన ఆకాశమంతసంతోషముందిలేరా

ప్రేమనుపంచినప్రేమనూ
ప్రేమనుపెంచినప్రేమనూ
ఆశగాకోరదాప్రతిహృదయం

ప్రేమనుపొందటంఒవరం
అదిఅంబరమంటినసంబరం
ప్రేమలోతేలుదాంప్రతినిమిషం
ప్రేమలోఈలోకమేసాగెప్రేమగా
ప్రేమతోఈజీవితంప్రేమించగా

-BOYINA SANTHOSH
ECE-B
Final year

తరిగేఆనందం

(Tarigeenanandam

పెరిగె కొద్దీ తరిగే ఆనందం
తొలి పొద్దులో గరిక పూవుపై
మంచు తాకి మైమరచింది నేనేనా?
ముంగిట ముగ్గుకి రంగులద్ది
మురిసిపోయిన మనిషి నేనేనా?

వాన చినుకుల్లో కలిసి తడిసి
అలిసిపోయిన మనసు నాదేనా?

రేకులు రాలుతున్న పూవును చూసి
చెక్కిలి జారిన కన్నీరు నాదేనా?
ఏది అప్పటి సున్నితత్వం?
ఏది అప్పటి భావుకత్వం?

-V.P.KALYAN KUMAR
ECE-B
Final year

నేస్తమా... (Buddies)

వేచే కోకిలకై వచ్చే వసంతంలా
నిరీక్షించే రైతులకై కురిసే వర్షంలా

ఎడారిలా వున్న నా జీవితంలో

ఎండమావి నువ్వయ్యావు
నీ ఆత్మీయ పలకరింపు నాకెంతో నచ్చింది

నీ మధురమైన మాటలతో నా మనసు

పులకరించింది

నువ్వు నా జీవితానికి స్ఫూర్తినిచ్చావు

జీవన గమనంలో అండగా నిలిచావు

ఈ తోడు శాస్త్రతంగా వుండాలి నేస్తం

ఏడు కాదు ఏడేడు జన్మలు నీదానిని, ఇది

వాస్తవం ...

-V.P.KALYAN KUMAR
ECE-B
Final year





KALPANA CHAWLA: THE INSPIRATION



Kalpana Chawla

“Just looking at Earth, looking at the stars during the night part of Earth; just looking at our planet roll by and the speed at which it goes by and the awe that it inspires; just so many such good thoughts come to your mind when you see all that, doing it again is like living a dream - a good dream - once again, ” said the first Indian woman to fly with a U.S. space shuttle crew, Kalpana Chawla

Kalpana Chawla was born in the city of Karnal in Haryana, in a home of three elder sisters and one elder brother. She as a student was hugely interested in science and always desired to make it into engineering and the science of space. Hence Kalpana took admission to DAV College for Women for her pre-University lessons. Later, she joined Dyal Singh College, just to pursue engineering from the illustrious Punjab Engineering College, Chandigarh.

Kalpana Chawla in NASA

It was precisely in NASA (National Aeronautics and Space Administration) that Kalpana Chawla lived her dream to travel and die in space. After completing her education in aeronautics, Kalpana received a call from the MCAT Institute, San Jose, California as a research scientist. The project was made under the guidance under NASA. For her research work in NASA, Kalpana had to study several complex features of earth and its gravity. Yet, she always yearned to fly and kept wishing on to finally get selected for a NASA mission to space.

Maiden Space Mission of Kalpana Chawla

Kalpana Chawla's maiden space mission was like an impossible dream coming true to her, owing to her severe perseverance and determination to make it big in aeronautics. Her first mission was to fly in Space Shuttle Columbia flight STS-87, which began in the late months of 1997. She was a part of the six-member international crew on board. Kept under strict discipline and control in the maiden mission, Kalpana had to monitor several functions of the planet, including zero gravitational activities. After initial minor malfunctions of Payload Bay, Kalpana's first mission was a huge success after touchdown. Overnight Kalpana Chawla had become a legend.

Final Space Mission of Kalpana Chawla

Kalpana Chawla's final space mission was in association with NASA yet again, on board the Space Shuttle Columbia, the one in which she made her first voyage, with flight STS-107. In spite of a first flight, she was most eager to resume a space flight, a rare among astronauts. Her mission this time however was purely scientific, with more complex work to be performed. Kalpana was assigned as the Flight Engineer and the Mission Specialist for STS-107. Her experiments in space would bring about path-breaking results for earth and its population. However, everything was cut short quite pathetically, when Columbia accidentally broke down just 16 minutes before landing, burning and annihilating everything and everyone with it on 1stFebruary,2003.

Awards of Kalpana Chawla

- Congressional Space Medal of Honour.
- NASA Distinguished Service Medal
- ASA Space Flight Medal.

*--T.M.Rajkumar,
Second Year-B*





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Sow a thought, and you reap an act;
Sow an act, and you reap a habit;
Sow a habit, and you reap a character;
Sow a character, and you reap a destiny

- Charles Reade



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- Magazine Team

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